

Thesis for degree of Doctor

Complex Phase Behaviors in Interacting Polymeric Mixtures: Thermodynamic Analyses and Molecular Dynamics Simulations

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(Abstract)

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Interacting polymeric mixtures have emerged as both a focal point and a versatile platform for designing stimuli-responsive materials, owing to the diversity and tunability of their intermolecular interactions. However, their phase behaviors are inherently complex, governed by the competition and coupling among multiscale interactions, which present significant challenges for theoretical interpretation and experimental prediction. To address this, this study investigates three representative systems—polymer/ionic liquid (IL) mixtures, polyelectrolyte solutions, and associating block copolymers—to elucidate the mechanisms by which multiscale interactions

regulate phase behaviors and drive microstructural evolution, thereby providing a systematic framework for the design of smart materials.

First, motivated by the remarkable thermal responsiveness of polymer/IL mixtures, experimental characterization and molecular dynamics (MD) simulations were combined to investigate the phase behavior of Polyvinylpyrrolidone (PVP) in 1-butyl-3-methylimidazolium tetrafluoroborate (BMIMBF₄). During heating, the films exhibited a transparent–opaque–transparent transition, analyzed by differential scanning calorimetry (DSC) and light scattering to examine the effects of temperature and polymer concentration. The system exhibited an unusual loop-type phase behavior, featuring the coexistence of both a lower critical solution temperature (LCST) and an upper critical solution temperature (UCST), surpassing conventional single thermal responsiveness in adaptability. To elucidate the underlying molecular mechanism, MD simulations using the OPLS5 all-atom force field were performed. Structural and cluster analyses revealed thermally induced conformational transitions of PVP chains from gradual collapse to re-extension, while free volume analysis further corroborated the loop-type phase behavior. Furthermore, by varying alkyl chain lengths of imidazolium-based ILs, the simulations revealed that PVP chains exhibit similar temperature-dependent conformational behaviors, indicating that this molecular feature effectively governs the loop-type phase behavior in PVP/ILs mixtures. These findings offer new insights into the molecular basis for designing thermoresponsive smart materials.

Second, in polyelectrolyte solutions, the interplay between electrostatic interactions and chain conformations results in rich and complex phase behaviors. To provide a more comprehensive description of oppositely charged polyelectrolyte blends, we constructed a Landau free energy based on the Random Phase Approximation (RPA) theory combined with a compressible equation of state for charged systems (Cho–Sanchez EOS). This framework incorporates excluded volume effects, chain connectivity, dispersion interactions, and long-range Coulomb forces. Within this model, we investigated both conventional order-disorder transitions and more complex ordering

transitions, spanning from fully charged to weakly charged systems. The analysis revealed the roles of chain-size asymmetry, charge effects, disparity in dispersion interactions, and pressure effects in shaping the phase behavior. Furthermore, we identified both barotropic and baroplastic responses of the ordering transitions, highlighting the potential of designing pressure-responsive polyelectrolyte materials. These results provide a mechanistic understanding for theoretical predictions and establish general principles for the rational design of polyelectrolyte systems.

Finally, the theoretical framework was extended to associating AB diblock copolymers. Based on the RPA theory of combined with the Cho–Sanchez EOS for associating systems, we constructed a Landau free energy functional that accounts for excluded volume, dispersion interactions, and the effective Flory interaction parameter (χ), defined as the sum of enthalpic (χ_H) and entropic (χ_S) contributions. The RPA analysis identifies the critical conditions and quantifies the effective interaction parameters. Representative self-consistent field theory (SCFT) density profiles illustrate the disordered and lamellar morphologies predicted by the framework. The resulting analysis elucidates the temperature- and pressure-dependent ordering of different diblock copolymers. It was found that the competition between enthalpic and entropic contributions gives rise to both cooling-induced ordering and the less conventional heating-induced ordering. Skewed phase boundaries in the phase diagrams underscore the role of dispersion interaction asymmetry in driving ordered morphologies. Furthermore, barotropic and baroplastic pressure responses were identified, and the effects of multibody interactions on phase boundaries were further explored, broadening our understanding of the self-assembly mechanisms in associating block copolymers. These results provide theoretical guidelines for the rational design of nanostructured materials with tailored thermoresponsive and pressure-sensitive properties.

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I. Introduction

1.1 Background

Stimuli-responsive materials,^{1–3} originally derived from early studies on environment-sensitive polymers,^{4–6} have evolved into functional systems capable of perceiving and adapting to external stimuli such as temperature, pH, light, electric fields, and ionic strength.^{7–12} Among them, interacting polymeric mixtures are particularly attractive owing to their structural designability, conformational flexibility, and tunable intermolecular interactions, which endow them with significant potential for applications in smart coatings, drug delivery, sensors, and flexible electronics.^{13–21} Their responsiveness arises from the stimulus-dependent competition and cooperation among multiple molecular interactions, including hydrogen bonding, electrostatic interactions, hydrophobic effects, and charge-transfer interactions.²² Poly(N-isopropylacrylamide) (PNIPAM) serves as a representative example, in which temperature-dependent hydrogen bonding and hydrophobic effects govern coil-to-globule transitions.^{13,23} Other notable examples in polyelectrolyte complexes formed through electrostatic attractions between polycations (PC) and polyanions (PA).^{24–28} In amphiphilic block copolymers, the interplay between hydrophilic and hydrophobic segments drives self-assembly into micelles, vesicles, and other nanostructures in aqueous media.^{29,30} Understanding how competing and cooperative interactions drive structural evolution and govern phase behavior has become a central issue in the study of stimuli-responsive materials.³¹ To that end, this work investigates the complex phase behaviors and underlying mechanisms of three representative systems: polymer/ionic liquid (IL) mixtures, polyelectrolyte solutions, and associating block copolymers.

Temperature, as one of the most common external stimuli, plays a particularly prominent role in regulating complex phase behaviors in polymeric mixtures.³² They may exhibit an upper critical solution temperature (UCST) behavior—phase separation

at low temperatures and miscibility at high temperatures—or a lower critical solution temperature (LCST) behavior, in which phase separation occurs upon heating.^{15,33–36} More complex dual thermal responses, such as loop-type (LCST–UCST) and hourglass-type (UCST–LCST) transitions, have also been reported (Figure 1-1).^{30,37–41} These phenomena typically arise from temperature-dependent inversions of competing molecular interactions, leading to opposite solubility trends in different temperature regimes. Compared with single thermal transitions, dual thermoresponsive systems offer enhanced adaptability and functional tunability.^{40,41} In addition to temperature, pressure serves as another important stimulus for tuning phase behavior. Pressure-induced responses are typically classified as barotropic or baroplastic, corresponding to cases where miscibility is suppressed or promoted upon pressurization (Figure 1-2).^{42–46} Understanding these pressure-driven mechanisms not only complements the thermodynamic framework established for thermal responses but also provides insight into the design of recyclable and energy-efficient polymer systems.^{46,47} In this study, experimentally, macroscopic phase transitions were characterized using differential scanning calorimetry (DSC) and temperature-dependent light scattering measurements;^{48,49} computationally, molecular dynamics (MD) simulations probed chain conformations and elucidated intermolecular interactions; and theoretically, based on mean-field theory mixing free energy landscape was quantified and the thermodynamic stability of the order-disorder transition (ODT) and order-order transition (OOT) among lamellar (LAM), hexagonal (HEX), and body-centered cubic (BCC) phases was evaluated.⁵⁰

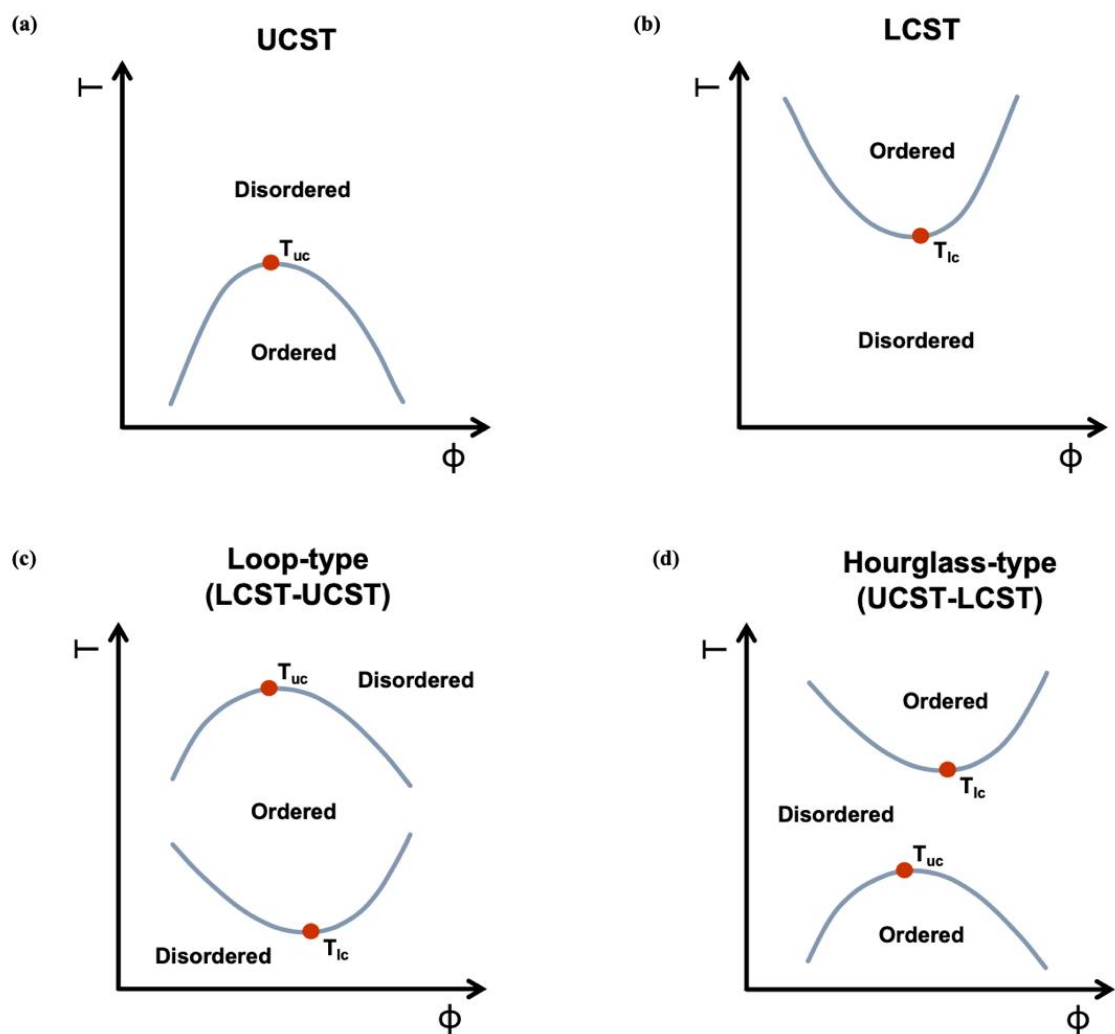


Figure 1-1. Four types of thermoresponsive phase transitions: (a) UCST (b) LCST (c) Loop (d) Hourglass. Blue lines represent the coexistence curve (binodal). T_{uc} and T_{lc} denote the upper critical and lower critical solution temperatures, respectively, where the ordered phase is below T_{uc} and above T_{lc} .

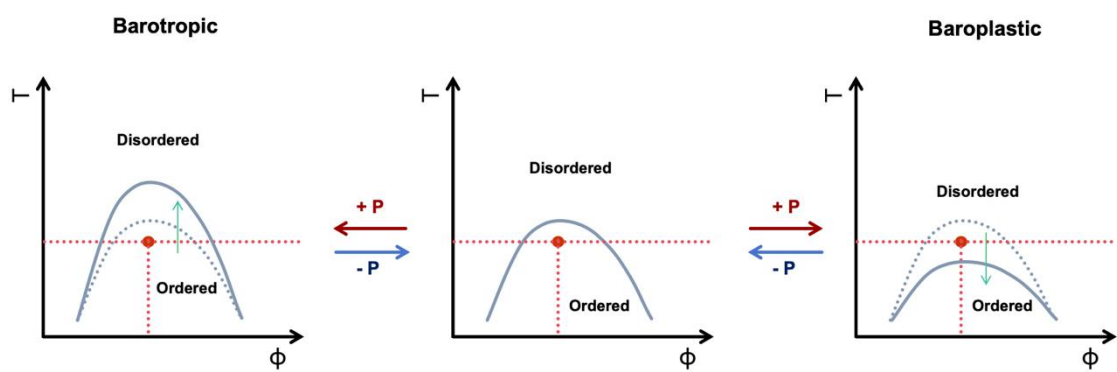


Figure 1-2. Schematic of pressure-responsive phase transitions with barotropic (left) and baroplastic (right).

1.2 Theoretical methods

Mean-field theories in polymer physics originated from Landau's classical framework, which provided a general description of phase transitions via a free-energy expansion in terms of an order parameter.⁵¹ Later, Edwards and de Gennes established the Gaussian thread model, which treats a polymer chain as an ideal continuous coil and provides the statistical basis for subsequent field-theoretic formulations.^{52–57} Building on this, Helfand developed the random phase approximation (RPA) to quantify small composition fluctuations in polymeric systems, enabling linearized predictions of spontaneous phase separation (spinodal) and ODT.^{58,59} Leibler combined RPA with a Landau expansion to extend it to symmetric diblock copolymers, thereby constructing the classical weak-segregation phase diagram.⁶⁰ Helfand and Fredrickson developed and generalized the self-consistent field theory (SCFT), providing a quantitative mean-field description of both ordered and disordered equilibrium morphologies by solving the field equations self-consistently.^{59,61–63} Building on these developments, Matsen developed numerical methods capable of accurately predicting the stability, morphology, and phase behavior of ordered structures in block copolymers under varying compositions and interaction parameters.^{64–66} Despite these advances, conventional field-theoretic methods generally assume incompressibility, limiting their ability to account for pressure effects. To overcome this limitation, Erukhimovich refined the theory by incorporating chain connectivity, pressure effects, and density fluctuations;^{67–70} Cho and Sanchez proposed a simplified compressible equation of state (EOS)^{71,72} based on the sticky hard-sphere chain (HSC) model of Baxter and Chiew^{73,74} combination with the Percus–Yevick approximation and mean-spherical approximation developed by Blum^{75,76}. In diblock copolymer systems, Cho further developed a compressible Landau approach, deriving the free energy either directly from A and B density fluctuations or through a transformation of the order parameters.^{77–79} These advances provide a consistent framework for analyzing phase separation, self-assembly, and pressure-responsive behaviors in polymeric systems.

Early theoretical descriptions of polyelectrolyte systems were established by the Voorn–Overbeek (VO) theory, which combined the Flory–Huggins mixing free energy with the Debye–Hückel electrostatic screening theory to describe phase separation in charged polymeric mixtures.^{80,81} However, the VO theory neglected the effects of chain connectivity and excluded volume, leading to discrepancies such as the incorrect prediction of salt partitioning.⁸² Subsequent Monte Carlo simulations by de Pablo⁸³ and bulk thermodynamic analyses by Wang^{84,85} confirmed that both chain connectivity and excluded volume are essential for accurately capturing the thermodynamic behavior of polyelectrolyte complexes. To that end, this study investigates the temperature- and pressure-dependent phase behaviors of polyelectrolyte systems, particularly complex coacervation, using a Landau expansion of the free energy based on RPA combined with the compressible Cho–Sanchez EOS.^{86,87} Building on the compressible Landau theory previously developed by Cho, the associating AB diblock copolymer system was reexamined by redefining the effective cubic order term to capture the coupling between density fluctuations and molecular interactions. The newly developed Landau free energy was derived in two equivalent ways: directly and via an order parameter transformation.

1.3 Molecule Dynamics simulations

The MD simulation originated from the pioneering work of Alder and Wainwright, who employed a square-well potential to study the time evolution of HS systems.⁸⁸ Their approach successfully captured the essential features of phase transitions in simple fluids. In 1964, Rahman extended MD simulations from idealized HS models to realistic liquids by employing the Lennard-Jones potential to describe liquid argon.⁸⁹ Verlet proposed a more efficient Verlet integration algorithm based on Rahman's work, establishing the numerical foundation for modern MD simulations.⁹⁰ In 1978, Rapaport studied MD simulations of long chain polymer with excluded volume in dilute solution, examining how the end-to-end distance (R_{ee}) and radius of gyration (R_g) scale with chain length, thus establishing the foundation for polymer MD studies.⁹¹ In recent years, improvements in simulation algorithms and the integration of force fields such as Optimized Potentials for Liquid Simulations-All Atom (OPLS-AA) have established MD simulation as an effective tool for probing polymer structural properties and elucidating phase behavior mechanisms.^{92–98} Accordingly, MD simulations of the polymer/IL systems in this study were carried out using the highly parameterized and accurate OPLS-5 force field^{94,95,99,100} implemented in the Schrödinger Maestro (2024-3) software. In this force field, the total potential energy of the system is expressed as:

$$E = E_a + E_b + E_d + E_i + E_{vdw} + E_{ele} \quad (1-1)$$

where E_a is angle bending energy, E_b is bond stretching energy, E_d is dihedral torsion energy, E_i is improper torsion energy, E_{vdw} is van der Waals (vdW) interaction energy, and E_{ele} is the electrostatic interaction energy.

II. Loop-type Phase Behavior in Polyvinylpyrrolidone/Ionic Liquid Mixtures

2.1 Introduction

ILs are molten salts composed solely of cations and anions that remain liquid at relatively low temperatures, including room temperature, and are considered “green” solvents due to their nonvolatility, high thermal stability, ionic conductivity, structural tunability, and excellent solvating properties.^{101–104} Owing to the limitations of conventional aqueous or organic solvents, polymer/ILs systems have attracted attention for the design of functional and responsive materials.^{21,105–107} Polyvinylpyrrolidone (PVP) is a water-soluble polymer known for its excellent solubility and biocompatibility.^{108,109} The presence of carbonyl groups (C=O) in its side chains allows it to interact favorably with ILs, making it an effective additive for enhancing ionic conductivity.^{110,111} While numerous studies have explored PVP/ILs systems,^{110–118} the specific phase behavior of the PVP/1-butyl-3-methylimidazolium tetrafluoroborate (BMIMBF₄) combination has remained insufficiently investigated. In this study, experimental characterization and MD simulation were combined to elucidate its thermoresponsive phase behavior. The films exhibited a unique loop-type phase behavior. To elucidate the underlying mechanism, MD simulations were performed using the OPLS-5 force field. Established partial charge parameters for all atoms of BMIMBF₄ were adopted from the literature to reasonably describe the equilibrium structural properties.^{94,95,119,120} This aimed to investigate the temperature-dependent conformational changes of PVP chains in BMIMBF₄, visualize phase separation phenomena, and systematically analyze the variation of free volume with temperature. Additionally, the influence of the alkyl chain length of imidazolium-based ILs on the conformational transitions of PVP chains was examined, and the corresponding thermally induced transitions were visualized using cluster analysis.

2.2 Methods

2.2.1 Experimental details and characterization

Polyvinylpyrrolidone (PVP, average $M_w \sim 1,300,000$) was purchased from Sigma-Aldrich (USA). A high M_w was deliberately chosen to facilitate the observation of macroscopic phase separation behavior. PVP was dried in a vacuum oven for 2 days at 60-80°C prior to further processing to eliminate hygroscopic moisture. The ionic liquid 1-butyl-3-methylimidazolium tetrafluoroborate (BMIMBF₄, $\geq 97.0\%$ (HPLC), Sigma-Aldrich, Switzerland) was purified using the activated alumina (Al₂O₃, Sigma-Aldrich) adsorption method. The adsorption column was further eluted with anhydrous acetonitrile to ensure the complete elution of BMIMBF₄ while leaving impurities on the alumina adsorption column. The collected solution was placed in a rotary evaporator to evaporate the solvent, followed by vacuum drying at 80-100°C for 2 days to eliminate residual solvent and moisture. PVP/BMIMBF₄ films were then prepared via the solution cast technique using anhydrous acetonitrile (Sigma-Aldrich) as the solvent. Specified amounts of PVP and BMIMBF₄ were dissolved in anhydrous acetonitrile to prepare solutions with polymer concentrations (C_{PVP}) of 40.78 wt%, 45.25 wt%, 49.79 wt%, 62.31 wt%, and 71.26 wt%. The resulting solutions were stirred with a magnetic stirrer for several hours to ensure complete homogenization. Finally, the mixed solutions were vacuum dried at 60-80°C for 2 days to remove residual solvent, yielding transparent and flexible films.

¹H nuclear magnetic resonance (NMR) spectroscopy, based on the interaction between nuclear spins and an external magnetic field, was employed to verify the purity and chemical structure. Measurements were conducted on a 400-MHz NMR spectrometer (JNM-ECS400, JEOL, Japan) using deuterated dimethyl sulfoxide-d₆ (DMSO-d₆) as the solvent. Chemical shifts were reported in parts per million (ppm). Differential scanning calorimetry (DSC) analyzes the thermal properties of materials by measuring the heat flow difference between a sample and a reference as a function of

temperature, as schematically illustrated in Figure 2-1. In this study, the thermal behavior of the films was characterized using a DSC instrument (Q20, TA Instruments, USA) under a nitrogen atmosphere at a heating rate of 5 °C/min. The transition temperatures of PVP/BMIMBF₄ films were characterized using light scattering measurement, as schematically illustrated in Figure 2-2. A vertically polarized He–Ne laser ($\lambda=632.8$ nm) was reflected by a mirror, transmitted through the sample, and passed through a horizontally oriented analyzer before being detected by a photodiode sensor. The samples were compression-molded into transparent films, and the transmitted light intensity was recorded as a function of temperature over the range of 30–230 °C at a heating rate of 5 °C/min.

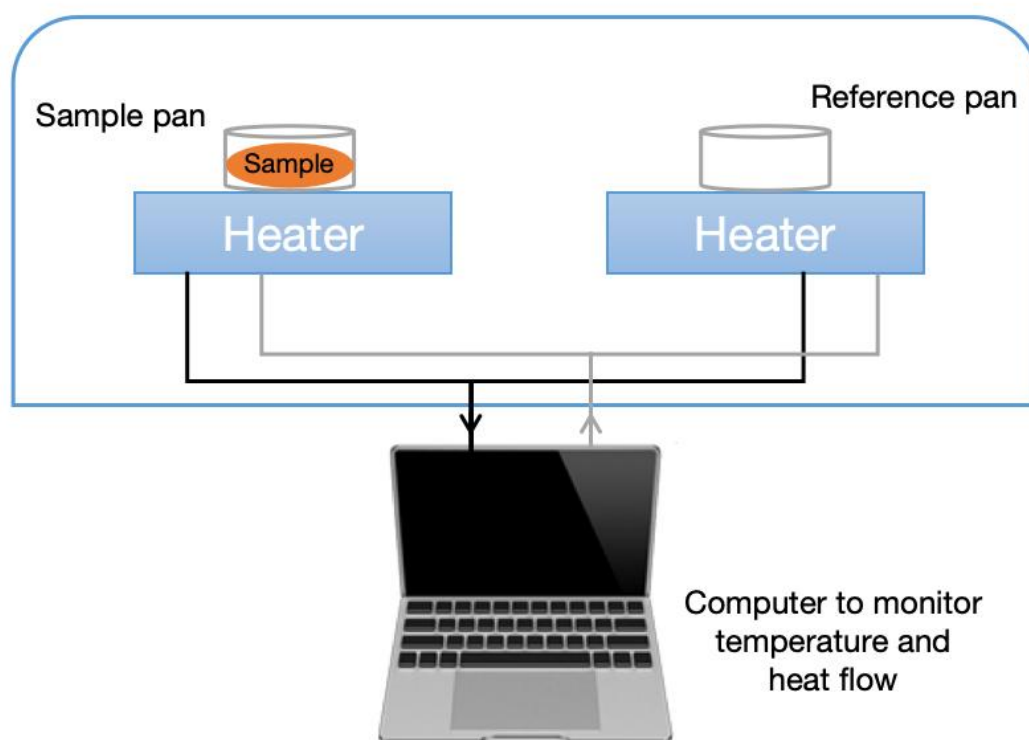


Figure 2-1. Schematic of the working principle of differential scanning calorimetry (DSC).

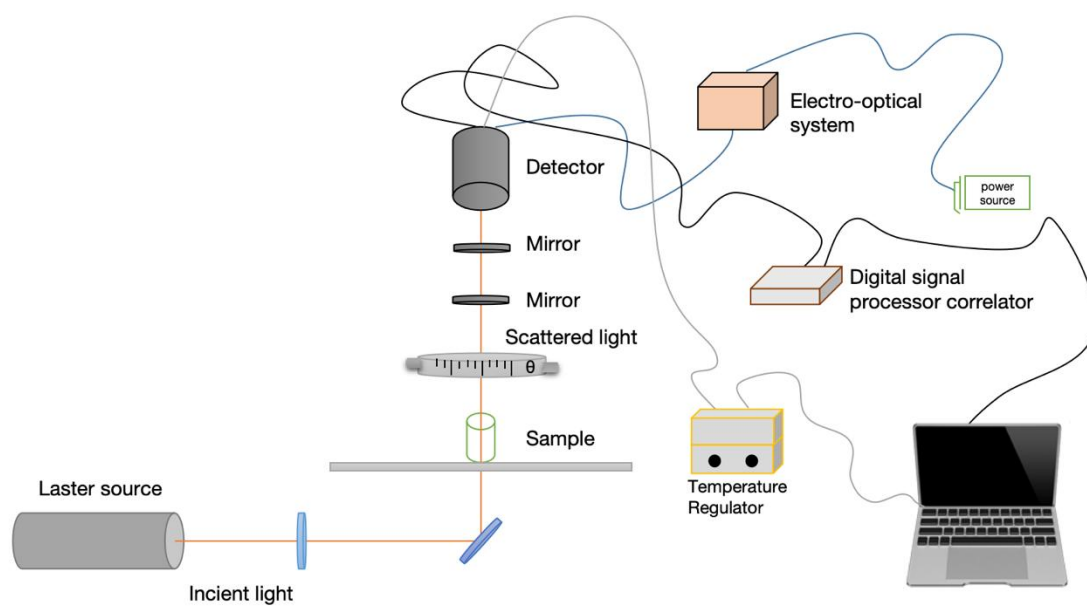


Figure 2-2. Schematic diagram of light scattering apparatus.

2.2.2 Molecular dynamics simulations

A disordered system was constructed in a cubic simulation box, consisting of 2 PVP chains with 50 repeat units each and 60 pairs of BMIM⁺ cations and BF₄⁻ anions as the initial configuration (Figure 2-3). Periodic boundary conditions were applied in all directions. The materials relaxation protocol (300 K, 1 atm) was applied, and the system annealing was under an NPT (constant number of molecules, pressure, and temperature) ensemble for a total simulation time of 2 ns to reach equilibrium. 200 ns NPT and NVT (constant number of molecules, volume, and temperature) ensemble simulations were carried out sequentially to ensure thermodynamic and configurational stability. The resulting equilibrium trajectories (Figure 2-4), free from thermal history effects, were employed for analyzing PVP₅₀ chain conformational changes and free volume. Table 2-1 lists the basic parameters of each system, including those with increased cationic alkyl chain lengths.

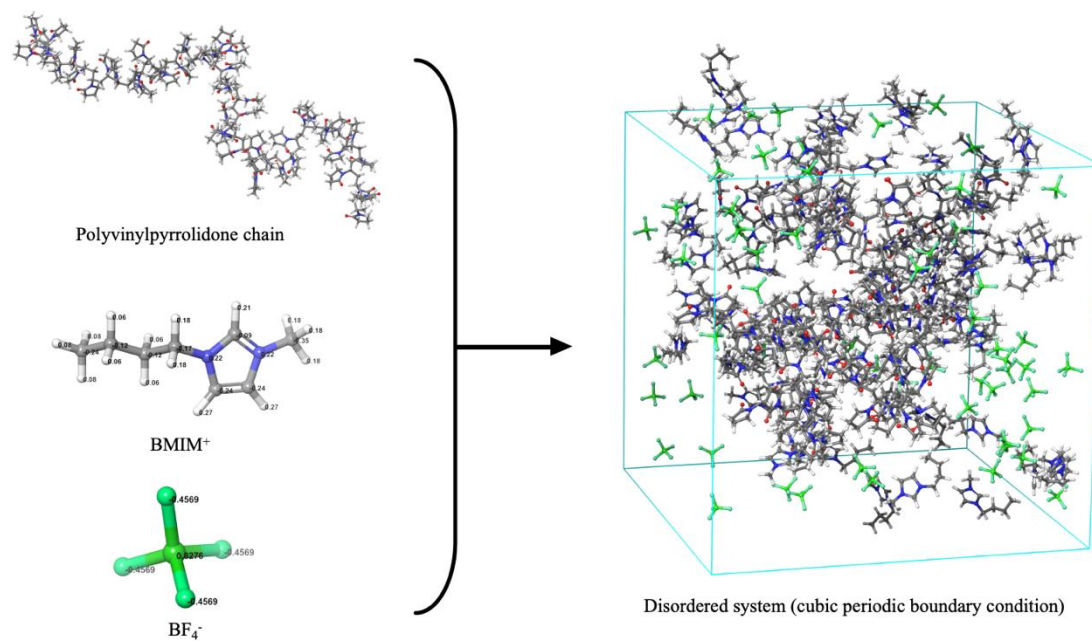


Figure 2-3. Initial configuration for molecular dynamics simulation. Structure of BMIMBF₄ with partial charges indicated on each atom. (gray atom is C, white atom is H, red atom is O, blue atom is N, green atom is B, faded green atom is F)

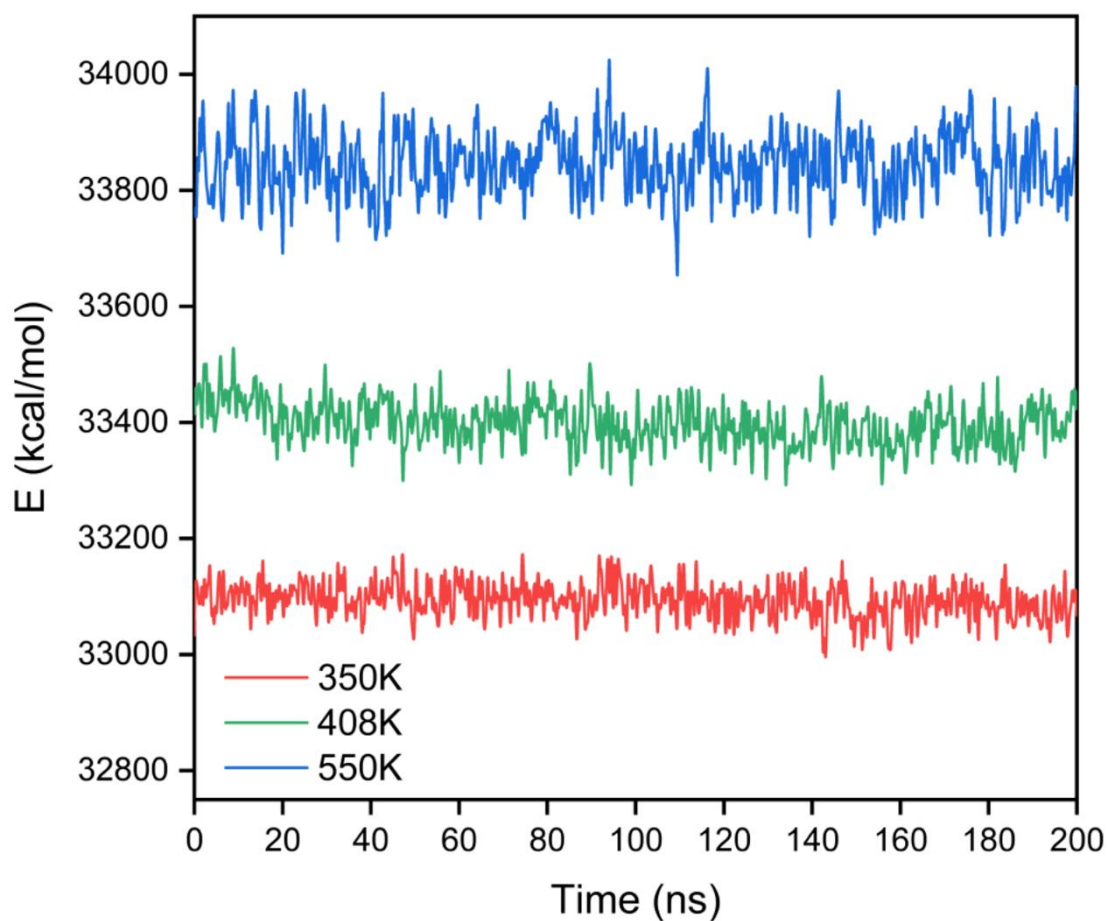


Figure 2-4. Time evolution of the total potential energy (E) at representative temperatures, showing stable plateaus indicative of equilibration. Similar behavior was observed at other temperatures.

Table 2-1. Basic information on each PVP/IL system.

Cell Component	C _{PVP} (wt%)	Cell size (Å ³)	MD simulated T (K)
2 PVP ₅₀ chains 60 BMIM ⁺ 60 BF ₄ ⁻	45	43.4452	350, 373, 383, 400, 408, 413, 418, 423, 433, 443, 453, 458, 463, 468, 473, 478, 483, 500, 523, 550, 600
2 PVP ₅₀ chains 60 HMIM ⁺ 60 BF ₄ ⁻	42.2	44.4109	350, 363, 373, 383, 388, 390, 393, 400, 410, 420, 423, 428, 433
2 PVP ₅₀ chains 60 OMIM ⁺ 60 BF ₄ ⁻	39.6	45.3362	350, 353, 363, 373, 378, 383, 388, 393, 400, 403, 405, 408, 413, 418, 423, 433, 443, 448, 450
2 PVP ₅₀ chains 60 DMIM ⁺ 60 BF ₄ ⁻	37.4	46.2253	350, 353, 358, 363, 368, 373, 378, 383, 385, 388, 390, 393, 395, 400, 413, 423

The conformational transitions of PVP₅₀ chains in ILs were quantitatively analyzed by computing the R_g , R_{ee} , and chain fractional anisotropy (FA). The R_g , which quantifies the size and compactness of the chains, was calculated according to the following equation:

$$R_g = \sqrt{\frac{\sum_{i=1}^N m_i |r_i - r_{cm}|^2}{\sum_{i=1}^N m_i}} \quad (2-1)$$

where N is the chain length, m_i is the mass of atom i , r_i is its position vector, and r_{cm} represents the position of the center of mass of the chain. The R_{ee} serves as an additional descriptor for characterizing polymer chain conformational changes. It was calculated by the following equation:

$$R_{ee} = \sqrt{2L_p L_e \left[1 - \frac{L_p}{L_e} + \frac{L_p}{L_e} e^{\frac{-L_e}{L_p}} \right]} \quad (2-2)$$

where L_p is the persistence length, and L_e is the extended chain length. The FA quantitatively describes the spatial orientation distribution of polymer chains, enabling the assessment of anisotropic behavior in PVP chains within the system. FA values were calculated for the PVP chains by the following equation:

$$FA = \sqrt{\frac{3}{2}} \cdot \sqrt{\frac{(\Lambda_1 - \bar{\Lambda})^2 + (\Lambda_2 - \bar{\Lambda})^2 + (\Lambda_3 - \bar{\Lambda})^2}{\Lambda_1^2 + \Lambda_2^2 + \Lambda_3^2}} \quad (2-3)$$

where Λ_1 , Λ_2 and Λ_3 are the eigenvalues of the gyration tensor of the polymer chain, and $\bar{\Lambda} = (\Lambda_1 + \Lambda_2 + \Lambda_3) / 3$ indicates the mean of the eigenvalues.

The free volume is a key structural parameter that characterizes the packing density and mobility of polymer chains. The fraction free volume (FFV) is defined as the fraction of the total system volume not occupied by atoms, calculated as:

$$FFV = \frac{V_{free}}{V_{system}} = 1 - \frac{V_{occ}}{V_{system}} \quad (2-4)$$

where V_{free} is the free volume, V_{occ} is the occupied volume, and V_{system} is the total system volume. The free volume of the system was evaluated using the probe method, in which a virtual probe moves along the vdW surface to generate the Connolly surface,^{121,122} thereby identifying interchain voids or gaps corresponding to unoccupied regions. The spatial distribution and size of these voids were determined using a grid-based method.

2.3 Results and discussion

2.3.1 Loop-type phase behavior of PVP/BMIMBF₄

¹H NMR spectroscopy was employed to analyze the purified BMIMBF₄. As shown in Figure 2-5, no detectable organic impurity peaks were observed, and the signal at ~3.3 ppm was attributed to residual water. To further investigate the chemical composition and potential interactions in PVP/BMIMBF₄ mixtures, ¹H NMR analysis was performed on a film with C_{PVP} of 45.25 wt% (Figure 2-6). The spectrum confirmed the presence of all characteristic peaks of both components and the absence of detectable impurities, indicating successful purification and reproducible film formation. Notably, the imidazolium C2–H proton exhibited a slight downfield shift ($\Delta\delta \approx +0.03$ ppm) relative to Figure 2-5, suggesting a weak interaction, likely hydrogen bonding, between BMIM⁺ and the C=O of PVP.

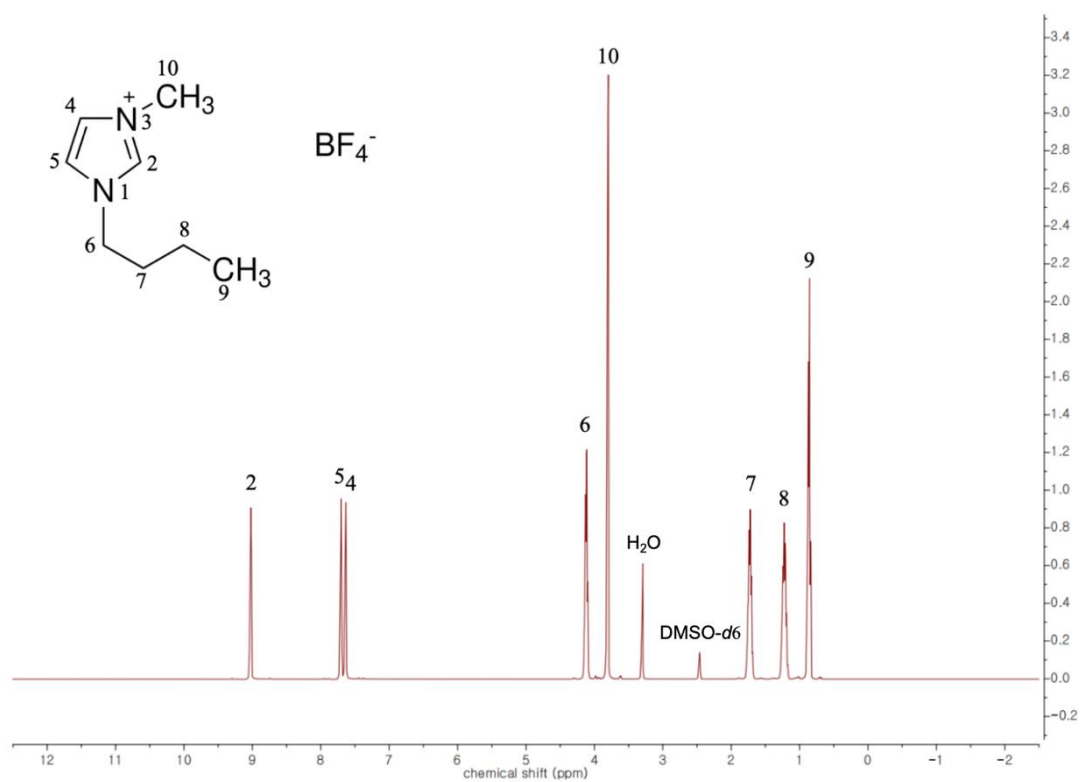


Figure 2-5. ^1H NMR spectrum (400 MHz) of BMIMBF₄ in DMSO-*d*₆.

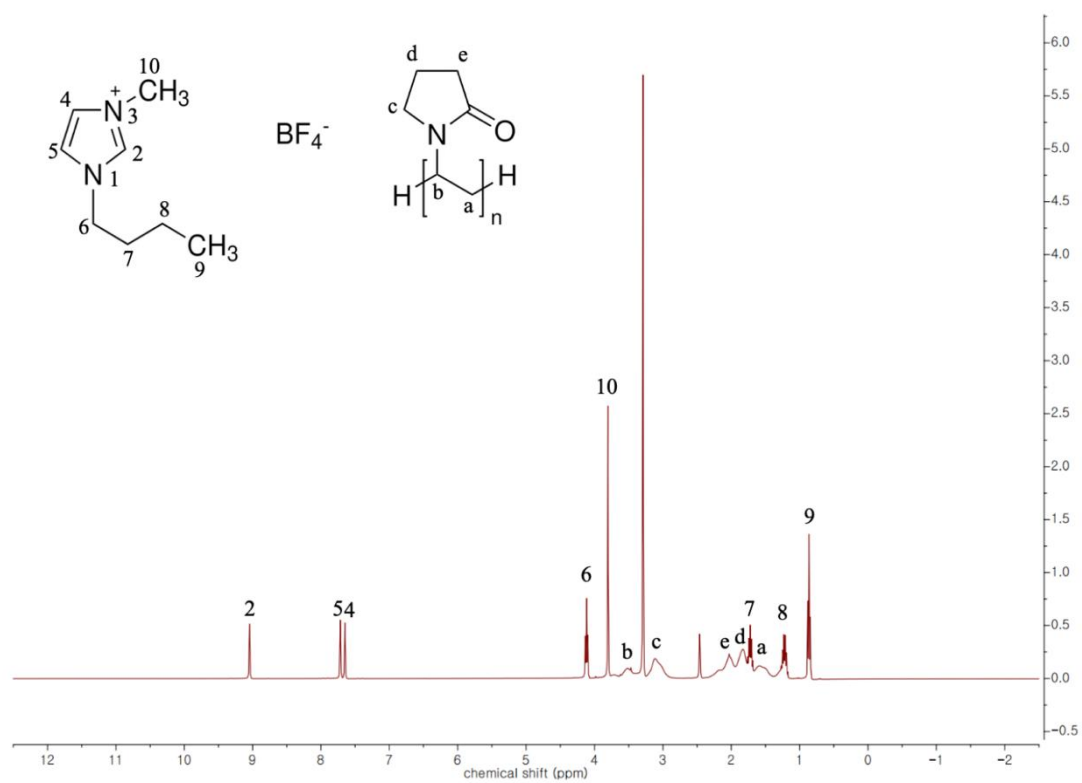


Figure 2-6. ^1H NMR spectrum (400 MHz) of $\text{C}_{\text{PVP}}=45.25$ wt% PVP/BMIMBF₄ film in DMSO-d₆.

A transparent–opaque–transparent transition was observed upon heating of PVP/BMIMBF₄ films within the studied concentration range. Figure 2-7 shows the representative macroscopic change observed for the film with C_{PVP} of 62.31 wt%, suggesting the presence of a loop-type thermoresponsive phase behavior. To further confirm this transition, the films were characterized using temperature-dependent light scattering measurements. Figure 2-8 exhibited the occurrence of the LCST–UCST transition for films with various C_{PVP}. At temperatures above the LCST, weakened polymer–solvent interactions combined with enhanced intrachain affinity lead to the collapse and aggregation of PVP chains, resulting in light scattering from locally enriched domains and a corresponding opaque appearance. Upon further heating beyond the UCST, increased thermal motion disrupts these domains, promoting the melting and dissolving of PVP crystallites into BMIMBF₄, thereby restoring optical transparency. This transition was also confirmed in the DSC thermograms (Figure 2-9). Although the melting peak of PVP was relatively weak in the DSC thermograms due to its amorphous nature, two distinct endothermic transitions were still observed, indicating thermally induced phase events. Figure 2-10 presents the phase diagram of PVP/BMIMBF₄ films, constructed by the endothermic peaks identified from DSC thermograms with the phase transition temperatures determined from light scattering measurements, revealing a characteristic loop-type thermal behavior. In the representative film with C_{PVP} = 45.25 wt%, the LCST and UCST were observed at ~135 °C and ~185 °C, respectively. This broad loop-type thermal window highlights the remarkable high-temperature adaptability of the IL-based polymer films. Moreover, variations in polymer concentration were found to systematically shift both transition temperatures, confirming the concentration-dependent nature of the phase behavior.

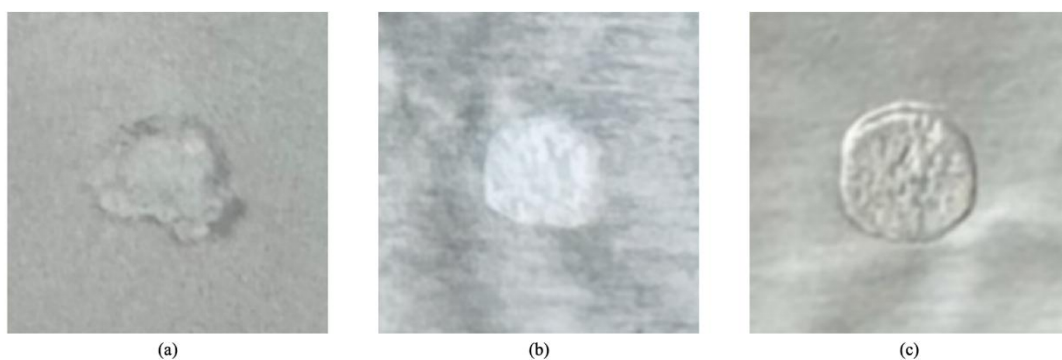


Figure 2-7. Photographic images of the PVP/BMIMBF₄ film with $C_{\text{PVP}}=62.31$ wt%, showing sample transparency changes upon heating on a hot stage. (a) Transparent state at room temperature. (b) Opaque state at $\sim 150^{\circ}\text{C}$. (c) Re-transparent state at $\sim 200^{\circ}\text{C}$.

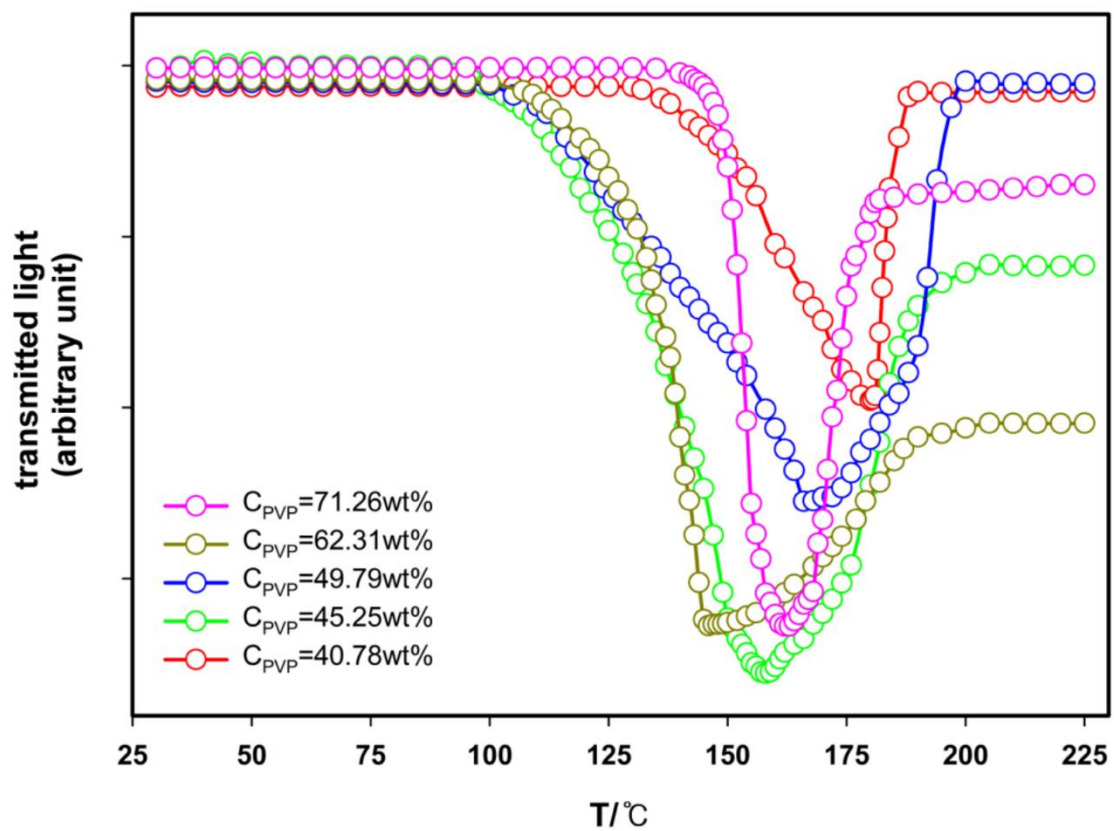


Figure 2-8. Light scattering measurements on PVP/BMIMBF₄ films for different C_{PVP} .

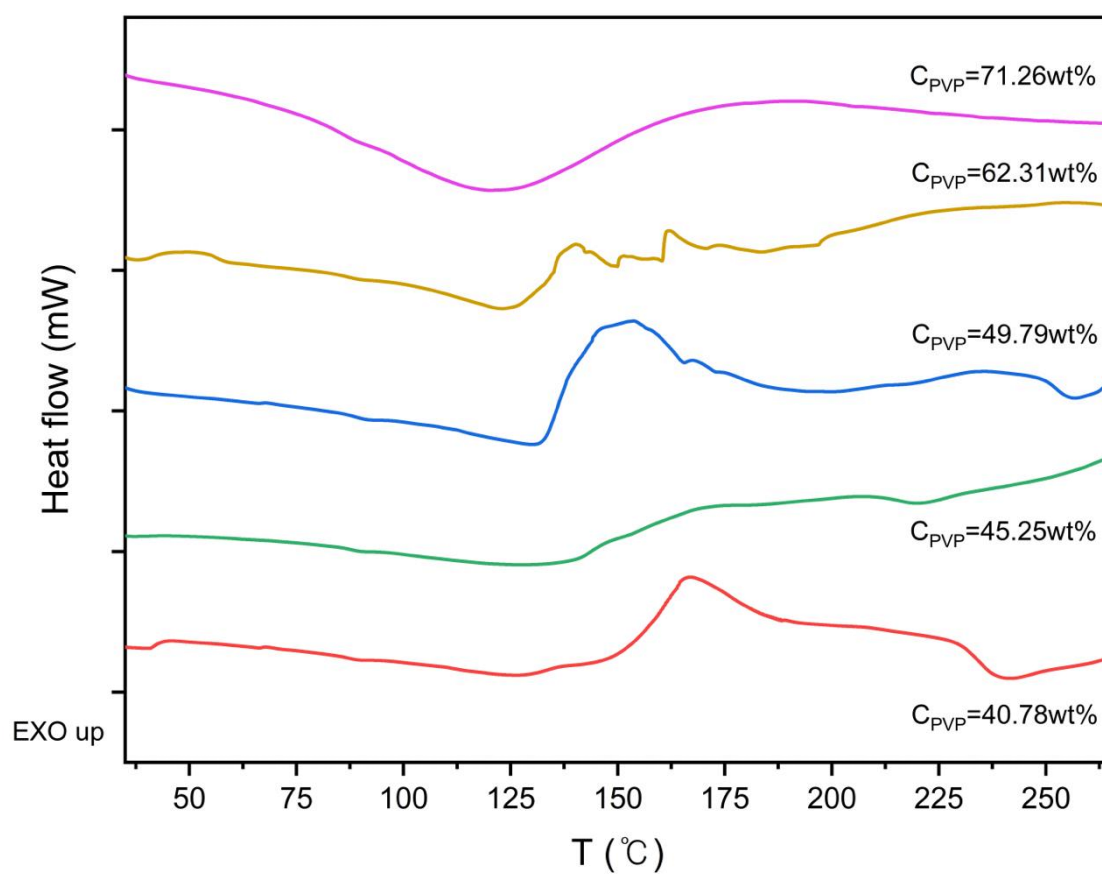


Figure 2-9. DSC thermograms of the melting temperature (T_m) of PVP/BMIMBF₄ films with different C_{PVP} .

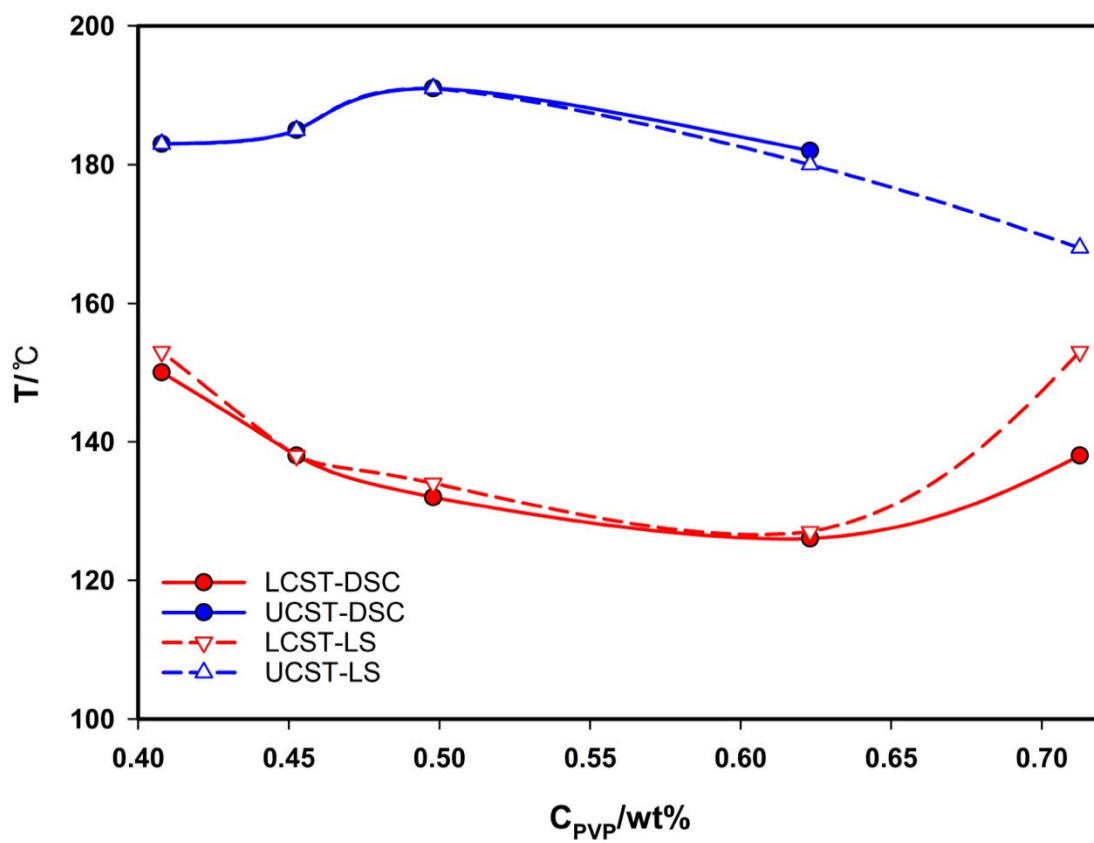


Figure 2-10. Loop-type phase behavior curves of PVP/BMIMBF₄ films.

2.3.2 Molecular dynamics simulation analysis of PVP chains conformations

Figure 2-11 (a) illustrates the time-dependent evolution of R_{ee} for PVP₅₀ chains evaluated at three representative temperatures. At 350 K, R_{ee} exhibits initial fluctuations but eventually stabilizes at approximately 26.2 Å. Upon heating to 408 K, R_{ee} rapidly decreases at around 68 ns and then stabilizes at approximately 18.1 Å. Upon further heating to 463 K, R_{ee} exhibits a sharp increase at around 105 ns, followed by stabilization at approximately 41.6 Å. Figure 2-11 (b) reflects the evolution of the spatial orientation of PVP₅₀ chains with time at three representative temperatures. At 350 K, the evolution of FA with time stabilizes at approximately 0.61. At 408 K, a lower FA value was recorded, showing a decrease from around 126 ns and eventually stabilizes at approximately 0.51. At 483 K, FA exhibits a rapid increase around 50 ns, ultimately stabilizing at approximately 0.78. The time evolution of R_{ee} and FA indicates that conformational transitions of the PVP₅₀ chain occur cooperatively upon reaching a critical structural threshold, involving rapid chain collapse or extension. Their eventual stabilization also confirms that the system achieved thermodynamic equilibrium.

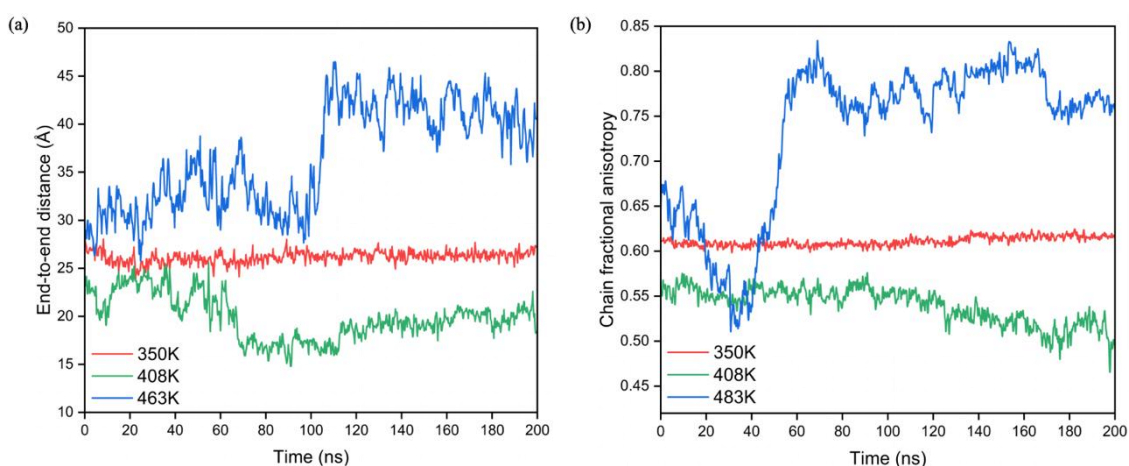


Figure 2-11. (a) Time evolution of end-to-end distance R_{ee} for PVP₅₀ chains at three different temperatures (350 K, 408 K, and 463 K); (b) Time evolution of FV for PVP₅₀ chains at three different temperatures (350 K, 408 K, and 483 K).

Figure 2-12 illustrates the temperature-dependent R_g curve for PVP₅₀, complemented by structural visualizations from cluster analysis. Figure 2-12 (a) depicts the conformational transition of the PVP₅₀ chains from an extended state to a clustered state, characterized by a pronounced decrease in R_g around 135 °C, indicating a gradual chain collapse over a temperature range. Conversely, Figure 2-12 (b) shows the conformational transition of PVP₅₀ from a collapsed, clustered state to an extended configuration, characterized by a sharp increase in R_g between 190 °C and 210 °C, followed by stabilization at a higher value. Cluster analysis was conducted using a 2.1 Å farthest-neighbor distance criterion, in which clusters within this threshold are rendered in the same color, and those beyond it are displayed in different colors. This analysis provides a clear visualization of the morphological evolution of the PVP₅₀ chains as the temperature increases, gradually transitioning from a dispersed state to the formation of clustered aggregates and back to a dispersed state.

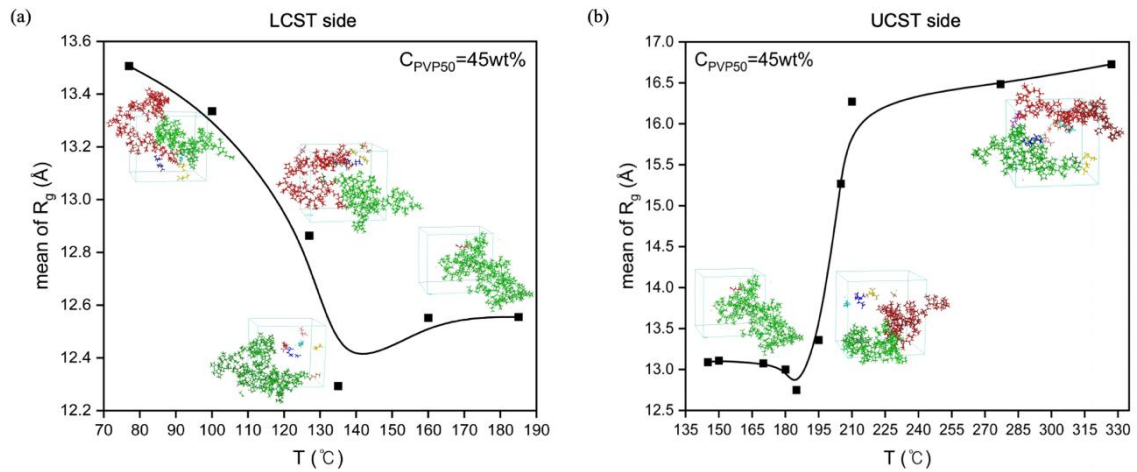


Figure 2-12. R_g for PVP₅₀ chains as a function of temperature, along with the visualizations of the structures on LCST (a) and UCST (b) sides. Green and red represent clusters containing PVP₅₀ chains.

Figure 2-13 (a) and (b) present the correlation between R_{ee} and temperature for the PVP₅₀ chains, showing good agreement with the R_g trends. Figure 2-13 (c) and (d) illustrate the temperature-dependent evolution of FA for the PVP₅₀ chains. Figure 2-13 (c) reveals a significant FA decrease within a temperature range, maintaining low values between 135 °C and 150 °C, reflecting a reduction in chain anisotropy and, in conjunction with the concurrent decrease trends in R_g and R_{ee} , indicating a transition toward a more compact, coiled conformation. Conversely, Figure 2-13 (d) reveals a sharp increase in FA within a higher temperature range, reaching a stable high value above 205 °C, indicative of enhanced chain anisotropy and the re-extension of PVP₅₀ chains. Overall, R_g , R_{ee} , and FA exhibit consistent trends, showing a sharp decrease near the LCST and a pronounced increase near the UCST, in excellent agreement with the experimental observations.

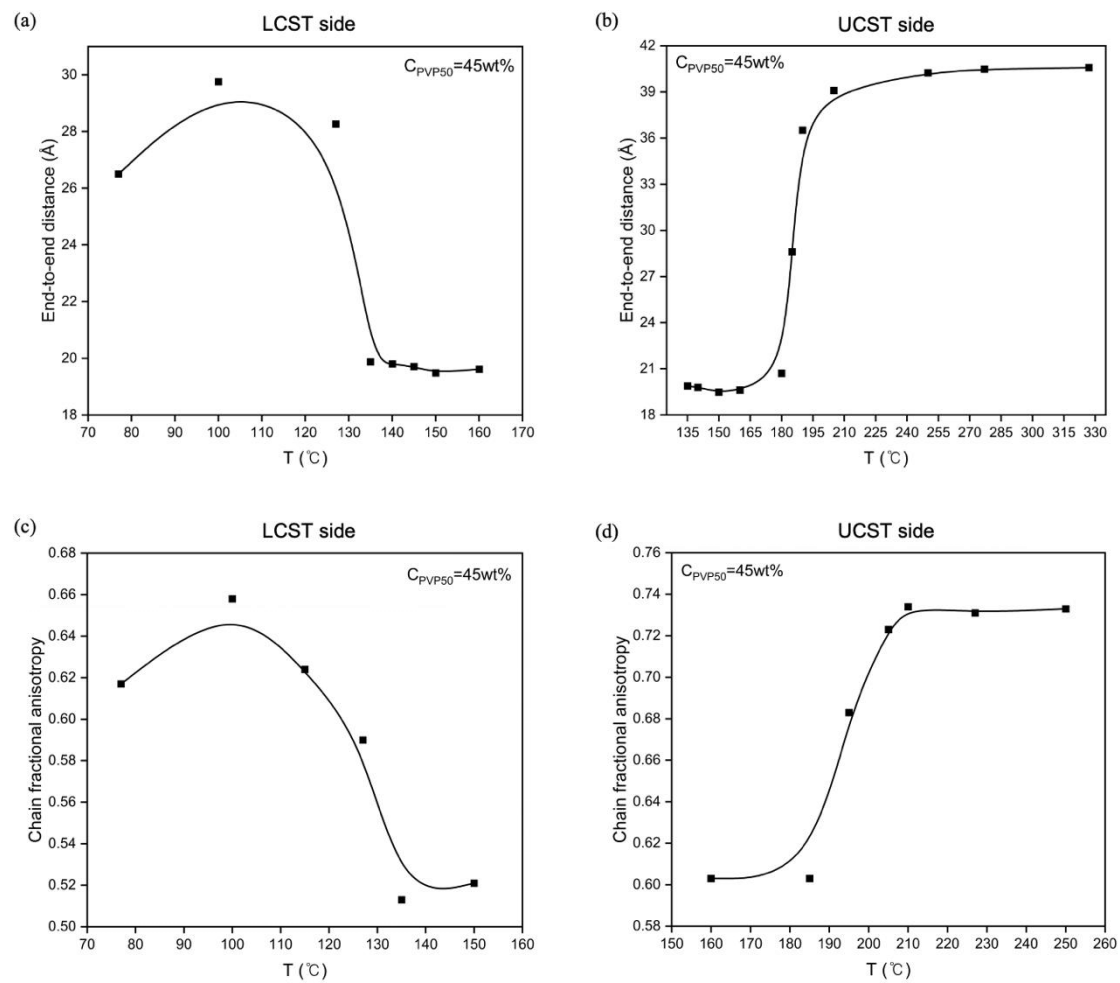


Figure 2-13. The end-to-end distance R_{ee} and chain fractional anisotropy FA for PVP₅₀ chains as a function of temperature. (a, b) show R_{ee} on the LCST and UCST sides, respectively; (c, d) show FA on the LCST and UCST sides, respectively.

To further elucidate the effect of alkyl chain length on the conformational transition and phase behavior, MD simulations were also performed for PVP/1-hexyl-, 1-octyl-, and 1-decyl-3-methylimidazolium tetrafluoroborate (HMIMBF₄, OMIMBF₄, and DMIMBF₄) systems. Figure 2-14 presents the temperature dependence of the R_g for each system, complemented by cluster analysis based on a farthest-neighbor distance criterion of 2.1 Å, which reveals similar conformational trends across all systems. Figure 2-15 presents the variation of LCST and UCST with the alkyl chain length, derived from the transition temperatures determined by both R_g and cluster analyses, respectively. The results reveal that increasing the alkyl chain length lowers the temperature at which phase separation occurs, as well as a reduction in the size of phase-separated domains. These further demonstrate that tuning the alkyl chain length provides an effective means to modulate the loop-type phase behavior in PVP/ILs mixtures.

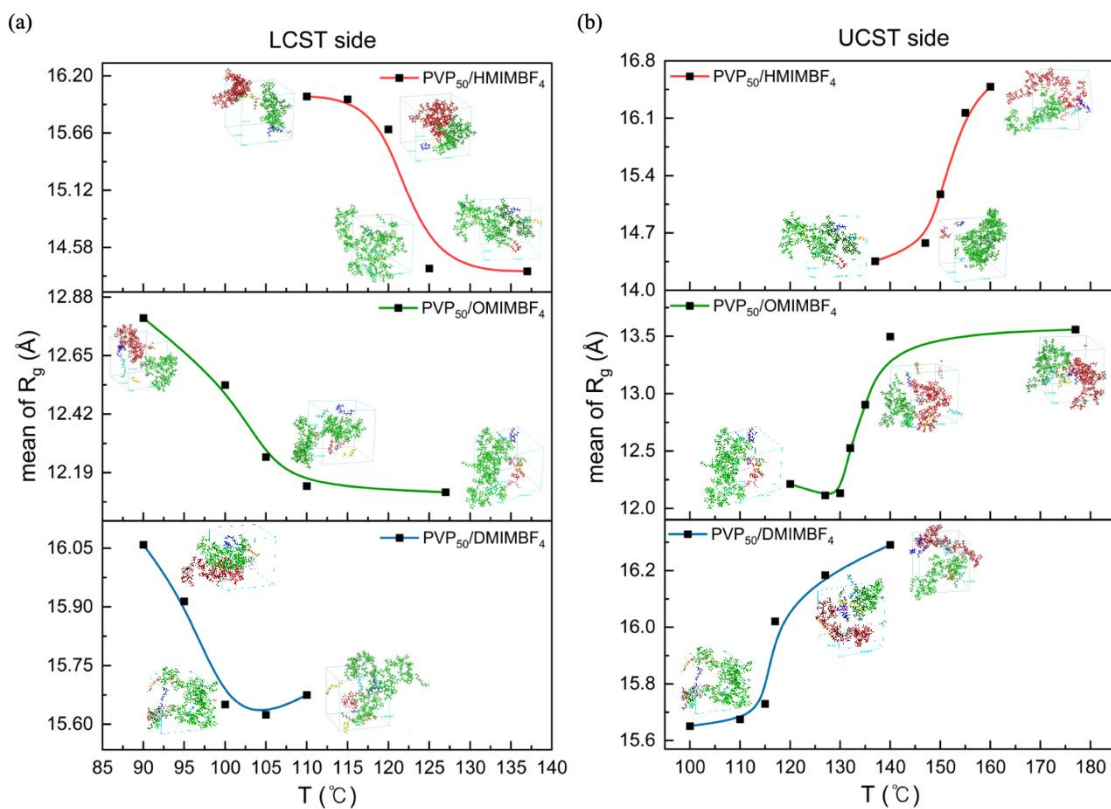


Figure 2-14. Temperature dependence of the R_g for PVP₅₀ chains in each system, along with the structural visualizations at the LCST (a) and UCST (b) sides. Green and red represent clusters containing PVP₅₀ chains.

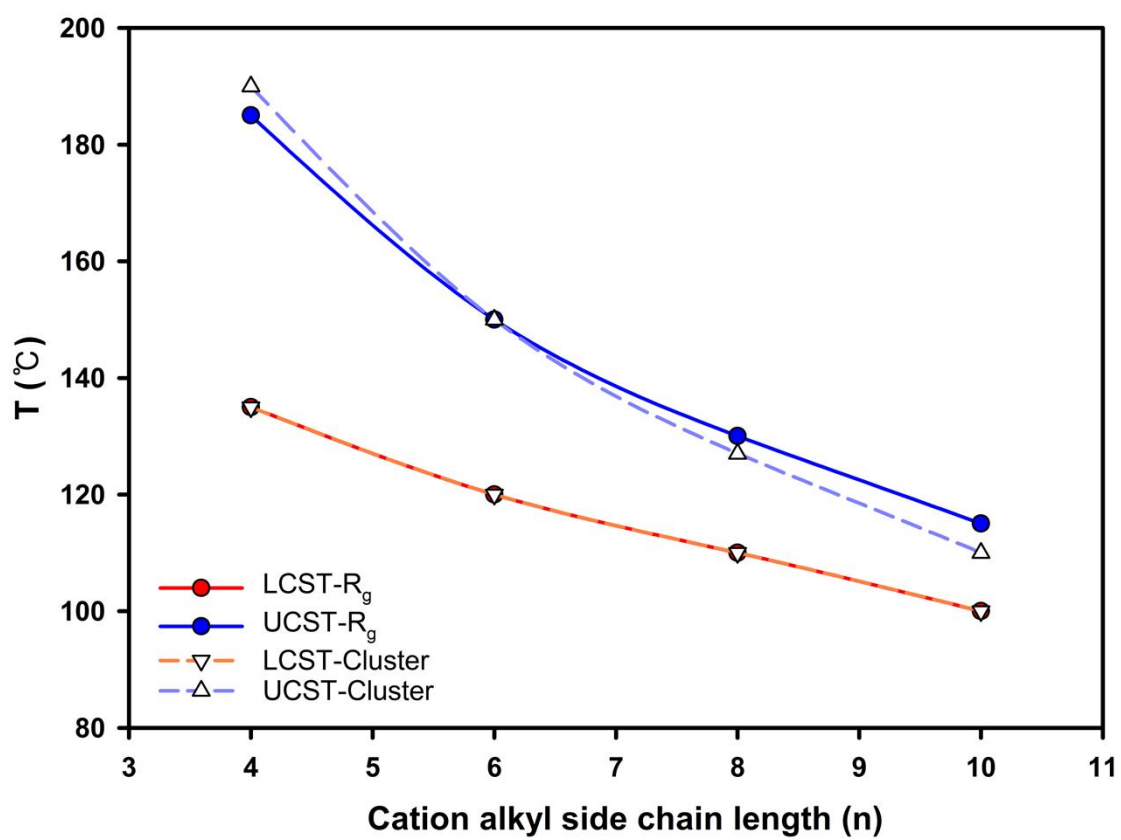


Figure 2-15. Dependence of LCST and UCST on alkyl chain length.

2.3.3 Molecular dynamics simulation analysis of free volume

Figure 2-16 displays the free volume morphology (blue areas) for varying probe radii at a constant temperature, revealing that the detected free volume increases as the probe radius decreases. For the computation of microvoids and free volume in the PVP/BMIMBF₄ system, the probe radius was set to 1.4 Å, corresponding to the effective molecular radius of water¹²³ and smaller than the vdW radii of the major polymer atoms ($r_C = 1.70$ Å, $r_N = 1.55$ Å, $r_O = 1.52$ Å)^{124,125}. This choice ensures that the probe can effectively access intermolecular voids while maintaining a physically realistic representation of the excluded volume. Figure 2-17 (a) and (b) illustrate the variation in the number of microvoids in the PVP/BMIMBF₄ system as a function of temperature. The number of microvoids was calculated across all simulation frames to ensure statistical robustness. As shown in Figure 2-17 (a), the microvoid count exhibits a sharp decline between 127 °C and 135 °C, reflecting a marked increase in molecular packing density. This abrupt decrease likely corresponds to a phase transition from a homogeneous to a phase-separated state. Remarkably, between 135 °C and 150 °C, the microvoid count remains consistently low, indicating sustained molecular compaction. This observation implies that strong intermolecular interactions maintain a densely packed structure despite further thermal input. Figure 2-17 (b) shows that below 185 °C, the microvoid count in the system remains minimal, consistent with polymer aggregation. A pronounced increase in microvoids occurs between 185 °C and 195 °C, marking a gradual transition from a phase-separated state to a homogeneous state, as polymer chains disperse. From 205 °C to 227 °C, the microvoid count stabilizes at a high level, indicating a fully homogeneous phase with enhanced chain extensibility. This temperature-dependent microvoid evolution quantitatively supports the loop-type phase behavior mechanism revealed by light scattering. Figure 2-17 (c) and (d) present the temperature-dependent evolution of free volume characteristics in the PVP/BMIMBF₄ system, with the left and right y-axes representing the FFV and

absolute free volume, respectively. In Figure 2-17 (c), below 127 °C, the system remains homogeneous; increasing temperature enhances molecular thermal motion, leading to system relaxation and increased local free volume. An inflection point between 127 °C and 140 °C signals the onset of intensified intermolecular interactions. Above 140 °C, phase separation induces abrupt FFV reduction, attributed to the densification of polymer-rich domains. In Figure 2-17 (d), persistently low FFV between 150 °C and 185 °C reflects a stable phase-segregated morphology. Between 185 °C and 195 °C, intensified thermal motion and weakened intermolecular interactions induce a sharp FFV increase, coinciding with the endothermic peak observed in DSC measurements, indicative of the transition from the phase-separated to homogeneous phase. From 205 °C to 227 °C, FFV stabilizes at a relatively high level, likely attributable to the reduced packing density of PVP₅₀ chains in the homogeneous phase.

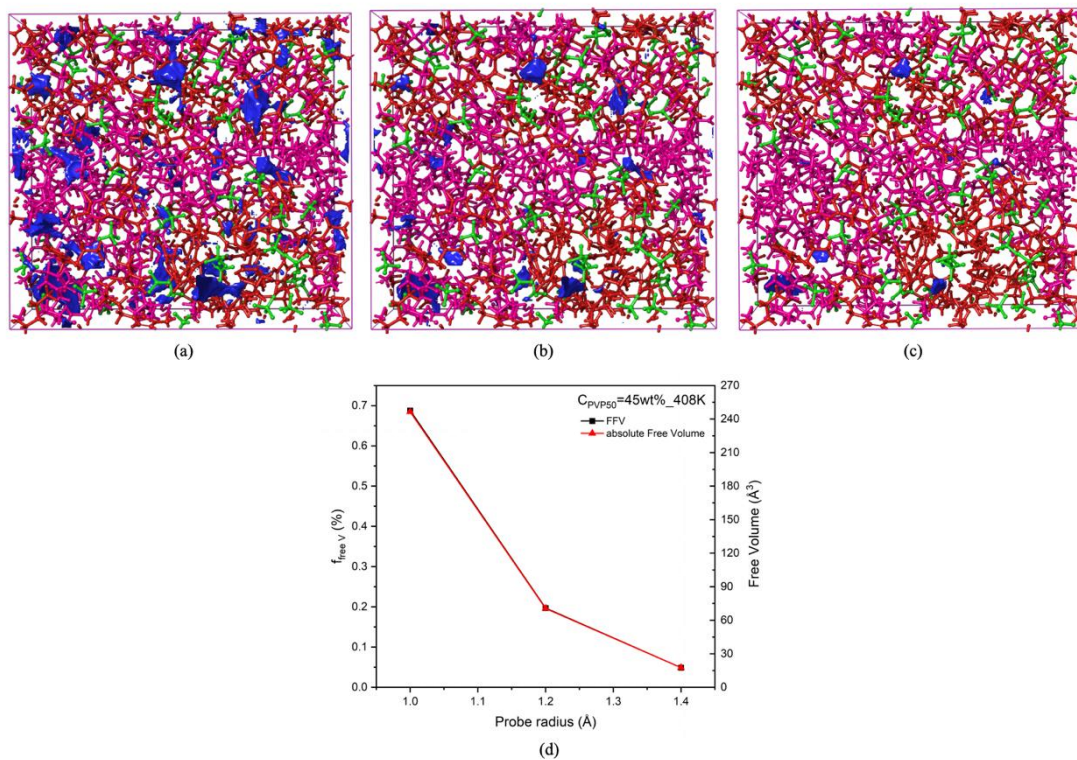


Figure 2-16. Free volume morphology (blue areas) of the system at 408K for different probe radii: (a) 1.0 Å; (b) 1.2 Å; (c) 1.4 Å; (d) FFV curve as a function of probe radius.

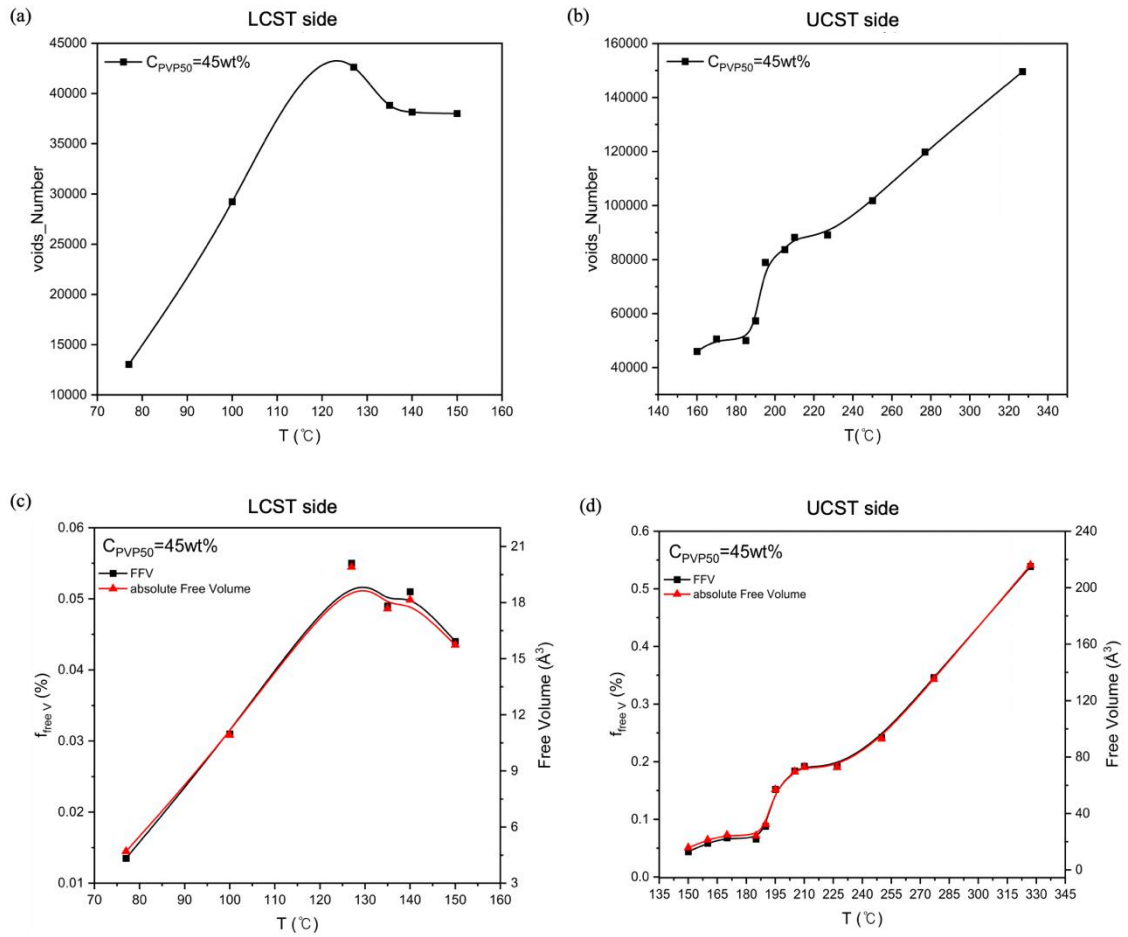


Figure 2-17. (a, b) Number of microvoids as a function of temperature on the LCST and UCST sides, respectively, at a probe radius of 1.4 Å. (c, d) Temperature-dependent evolution of free volume characteristics on the LCST and UCST sides, respectively, at the same probe radius.

2.4 Summary

This study investigated the thermoresponsive phase behavior of PVP/BMIMBF₄ films over a wide range of polymer concentrations. Light scattering measurements and DSC analyses revealed a loop-type phase behavior in this system. The resulting broad thermal response window, reflecting enhanced adaptability at elevated temperatures, highlights potential applications in thermal sensing and actuation. MD simulations further elucidated the microscopic mechanism, showing thermally induced conformational transitions of PVP chains that gradually collapse into compact clusters and re-extend at higher temperatures. These transitions correlate with non-monotonic variations in FFV, underscoring the competition between conformational entropy and polymer–solvent packing interactions. Furthermore, the study of the effect of alkyl chain length on phase behavior revealed that longer chains lead to a decrease in phase separation temperature and a contraction of the two-phase region. Collectively, these experimental and computational findings provide fundamental insight into the thermoresponsive nature of PVP/ILs mixtures and establish a framework for the rational design of smart materials with tunable phase behavior.

III. Complex Phase Behaviors in Polyelectrolyte Solutions

3.1 Introduction

Polyelectrolytes are polymers bearing ionizable groups along their molecular chains, which partially or fully dissociate in solution to form charged polymeric chains.^{126,127} Naturally occurring biomacromolecules such as nucleic acids, proteins, and polysaccharides inherently exhibit polyelectrolytic characteristics due to their charged groups, whereas synthetic polymers can acquire similar properties through chemical modification.^{127,128} Of particular interest are oppositely charged polyelectrolyte solution systems, which spontaneously undergo complex coacervation and microphase separation driven by electrostatic attraction. These coacervates hold significant potential for applications such as drug delivery and stimuli-responsive hydrogels.^{16,126,129,130} However, the underlying mechanisms remain incompletely understood, thereby limiting the accurate prediction of critical conditions in material design. A recent report¹³¹ of observed LCST phase behavior in complex coacervates of poly(styrene sulfonate) (KPSS)/poly(diallyl dimethylammonium) bromide (PDADMA) with added KBr has prompted a re-evaluation of the prevailing theoretical frameworks. Given the recognized importance of chain connectivity and excluded volume effects for accurately capturing the thermodynamic behavior of polyelectrolyte complexes, and considering that existing theories generally assume incompressibility, which limits their applicability.^{83,85,132} To address this, we employ a newly developed Landau analysis method that integrates the RPA with a compressible Cho–Sanchez EOS, systematically investigating the effects of topological structures of polyelectrolytes, including linear and ring architectures, dispersion interactions, chain-size asymmetry, and charge distribution on the complex phase behavior and pressure response of oppositely charged polyelectrolyte solution systems.

3.2 Theoretical methods

A staged modeling approach is employed to elucidate the electrostatic interactions and thermodynamic behavior of polyelectrolyte solutions, starting with a fully charged system comprising PA, counterions (C), and uncharged solvent (S), and subsequently considering a weakly charged blend system of PA and PC. Our system occupies a total volume V with an overall packing density η and is modeled as HSC, within a dielectric continuum of permittivity ε , further simplified by assuming that all monomeric segments are tangent hard spheres of identical diameter σ . Each polymer chain of species k consists of N_k rigidly connected spheres, and there are n_k chains. The close packed volume of g -species chains is given by $V_k = n_k N_k \nu^*$, where $\nu^* = \pi\sigma^3/6$ is the monomeric volume. The corresponding close-packed volume fraction is $\phi_k = V_k / \sum_l V_l$. The effective packing density of each component is therefore $\eta_k = \phi_k \eta$. Each monomeric unit may carry an electric charge $f_k z_k e$, where e is the elementary charge, $z_k = \pm 1$ is the charge number, with only a fraction f_k of the segments being ionized on average. The global electroneutrality of the system is maintained by requiring $\sum_k \phi_k f_k z_k = -\phi_A f_A + \phi_C f_C = 0$. Any pair of nonbonded g - and k -monomers separated by a distance r interact through a potential consisting of excluded volume and Coulombic interactions:

$$U_{gk}(r) = \begin{cases} \infty, & r < \sigma \\ \frac{f_g z_g f_k z_k e^2}{4\pi \varepsilon r}, & r \geq \sigma \end{cases} \quad (3-1)$$

The Helmholtz free energy A of the system is expressed using Cho–Sanchez EOS as the sum of ideal A_{id} and nonideal A_{ni} contributions.^{79,86,87,133} The nonideal free energy A_{ni} is further decomposed as $A_{ni} = A_{HSC} + A_{ele} + A_{ele-pol}$,

where A_{HSC} represents the excluded volume contribution by uncharged hard-sphere chains, A_{ele} electrostatic interactions of unconnected charged hard spheres, $A_{ele-pol}$ the chain connectivity contributions due to ion-ion correlations. The ideal chain mixing free energy is expressed as follows:

$$\frac{\beta A_{id} \nu^*}{V} = \sum_k \frac{\eta_k}{N_k} \ln \left(\frac{\eta_k T_k}{N_k} \right) \quad (3-2)$$

where $\beta = 1/\kappa_B T$, T_k is a molecular constant that is not relevant to thermodynamic properties. According to Chiew's connectivity of Baxter's adhesive hard spheres integral equation,^{73,74} the expression of HSC mixing free energy is:

$$\frac{\beta A_{HSC} \nu^*}{V} = \frac{3}{2} \left[\frac{\eta}{(1-\eta)^2} - \left(1 - \frac{1}{N} \right) \frac{\eta}{1-\eta} \right] - \frac{\eta}{N} \left[\ln(1-\eta) + \frac{3}{2} \right] \quad (3-3)$$

Based on the MSA theory,^{75,76} the expression for A_{ele} is as follows:

$$\frac{\beta A_{ele} \nu^*}{V} = - \frac{L_B}{\sigma} \sum_k \eta_k (f_k z_k)^2 \left(\frac{\Gamma \sigma}{1 + \Gamma \sigma} \right) + \frac{(\Gamma \sigma)^3}{18} \quad (3-4)$$

where $L_B = e^2 \beta / 4\pi \epsilon$ is the Bjerrum length to measure the strength of the ionic environment, and Γ is the shielding parameter. The dispersion interaction is expressed in mean-field energy as follows:

$$U_{nb-disp} = \frac{V}{2\nu^*} \sum_{gk} \phi_g \phi_k \bar{\epsilon}_{gk} u(\eta) \eta \quad (3-5)$$

where $\bar{\epsilon}_{gk}$ is the characteristic gk -sphere contact energy, $u(\eta)$ gives the density dependence of the energy as $u(\eta) = 4 \left[\left(\frac{\gamma}{C} \right)^4 \eta^4 - \left(\frac{\gamma}{C} \right)^2 \eta^2 \right]$, with $\gamma = \frac{1}{\sqrt{2}}$ and $C =$

$$\frac{\pi}{6}.$$

The free energy density functional for inhomogeneous charged chain mixtures is expressed as follows:

$$\begin{aligned} \frac{\beta A_{inhm} \nu^*}{V} = & \sum_k \frac{\eta_k}{N_k} \ln(\eta_k T_k) - \sum_k \frac{\eta_k}{N_k} \ln \left(\frac{1}{V} \int d\vec{r} \cdot q_k(\vec{r}, l) \right) + \frac{1}{V} \left(\int d\vec{r} \cdot \right. \\ & \left. \beta a_{ni}(\vec{r}) \nu^* + \int d\vec{r} \cdot \left\{ \eta_e(\vec{r}) e \beta \Psi(\vec{r}) - \frac{\beta \epsilon \nu^*}{2} |\nabla \Psi(\vec{r})|^2 \right\} - \right. \\ & \left. \sum_k \int d\vec{r} \cdot g \omega_k(\vec{r}) \cdot \eta_k(\vec{r}) \right) \end{aligned} \quad (3-6)$$

where q is the end-segment distribution function, $a_{ni}(\vec{r})$ is the localized version of the excess free energy density a_{ni} to give the effective short-ranged interactions. The long-ranged electrostatic energy contributions to the free energy functional are described by an electric potential $\Psi(\vec{r})$ that is generated by the charges present with the charge density $\eta_e(\vec{r}) = z_A \eta_A(\vec{r}) + z_C \eta_C(\vec{r})$. The vertex functions were formulated by considering the Gaussian part, the local interaction, and the long-ranged Coulombic interaction as follows:

$$\left[\Gamma_{gk}(\vec{q}) \right] = \left[S_{gk}^{0-1}(\vec{q}) \right] + \left[\frac{\partial^2 \beta a_{ni} \nu^*}{\partial \eta_g \eta_k} \right] + \left[\frac{z_g z_k f_g f_k \beta}{\epsilon \nu^*} \cdot \frac{1}{q^2} \right] \quad (3-7)$$

The parametrized Landau free energy takes the following form:

$$\begin{aligned} \beta \Delta A = & \beta [A_{inhm} - A_{hm}] \\ \approx & \left[\Gamma_{AA} \zeta_{n,A}^2 - 2\Gamma_{AC} \zeta_{n,A} \zeta_{n,C} + \Gamma_{CC} \zeta_{n,C}^2 \right] - \left| \alpha_{n,A} \zeta_{n,A}^3 - \alpha_{n,C} \zeta_{n,C}^3 \right| + \delta_{n,A} \zeta_{n,A}^4 \\ & + \delta_{n,C} \zeta_{n,C}^4 \end{aligned} \quad (3-8)$$

where ΔA is the difference between inhomogeneous and homogeneous free energies. $\zeta_{n,g}$ is amplitude for g -density fluctuations, $\alpha_{n,g}$ is the cubic coefficient, and $\delta_{n,g}$ is the quartic coefficient, respectively, given for LAM, HEX, and BCC as:

$$\delta_{n,g}^{LAM} = \frac{3!}{4!} \cdot \Gamma_{gggg}(0, 0) \quad (3-9-1)$$

$$\delta_{n,g}^{HEX} = \frac{18}{4! 3^2} \cdot [\Gamma_{gggg}(0, 0) + 4\Gamma_{gggg}(0, 1)] \quad (3-9-2)$$

$$\delta_{n,g}^{BCC} = \frac{36}{4! 6^2} \cdot [\Gamma_{gggg}(0, 0) + 8\Gamma_{gggg}(0, 1) + 2\Gamma_{gggg}(0, 2) + 4\Gamma_{gggg}(1, 2)] \quad (3-9-3)$$

3.3 Results and discussion

3.3.1 Fully charged system

The topological effect on the phase behavior was investigated using two representative models of PA, linear (L) and ring (R), each composed of $N_{PA} = 50$ and $N_C = N_S = 1$ in a fully charged system (Figure 3-1). The analysis began with the bulk thermodynamics of the system without vdW attractions to determine the scope of phase separation. The results revealed the coexistence of a PA-rich phase and an almost pure solvent phase, which arises from the strong electrostatic shielding effect. Figure 3-2 compares the bulk thermodynamic analyses of the L and R systems. Notably, the shielding effect appears to be stronger in the R than in the L counterpart. Proceeding from the bulk thermodynamic analysis to the RPA analysis, Figure 3-3 compares the L and R systems at a dielectric constant of $\epsilon_r = 25$. The macrophase critical point (CP) appears at $\phi_A = 0.01625$ for the L system and $\phi_A = 0.01624$ for the R system. The onset of microphase separation occurs at $\phi_A = 0.0264$ for the L system and at $\phi_A = 0.052$ for the R system. These results indicate that the macrophase CP becomes aphysical due to shielding-induced microphase separation. Furthermore, the macrophase CP shifts toward the PA-rich region for the R system, and the microphase region becomes broader for the R system than for the L system.

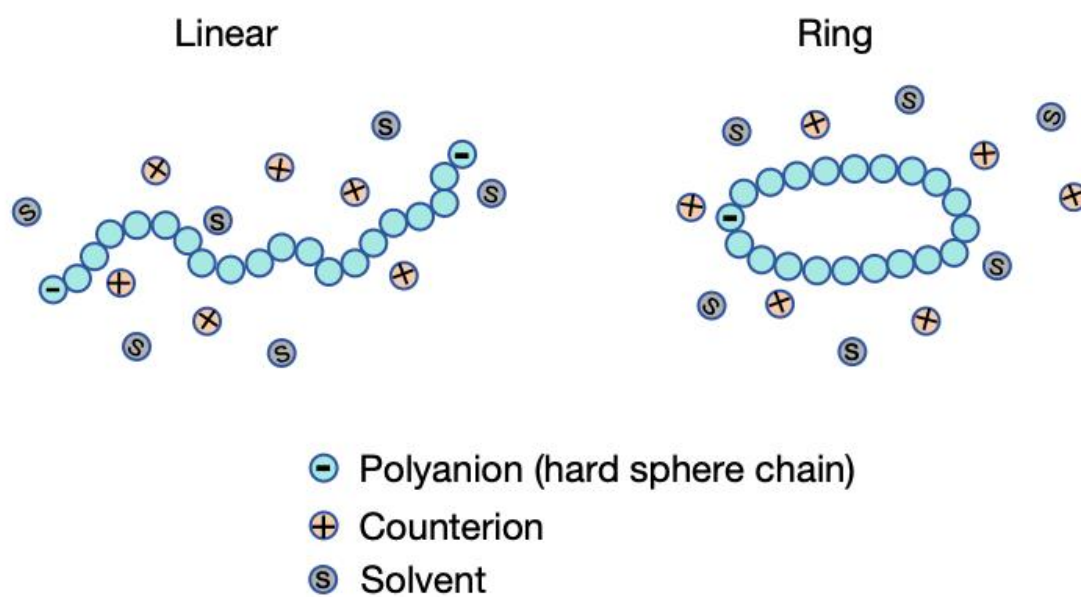


Figure 3-1. Schematic diagram of linear (L) and ring (R) model systems.

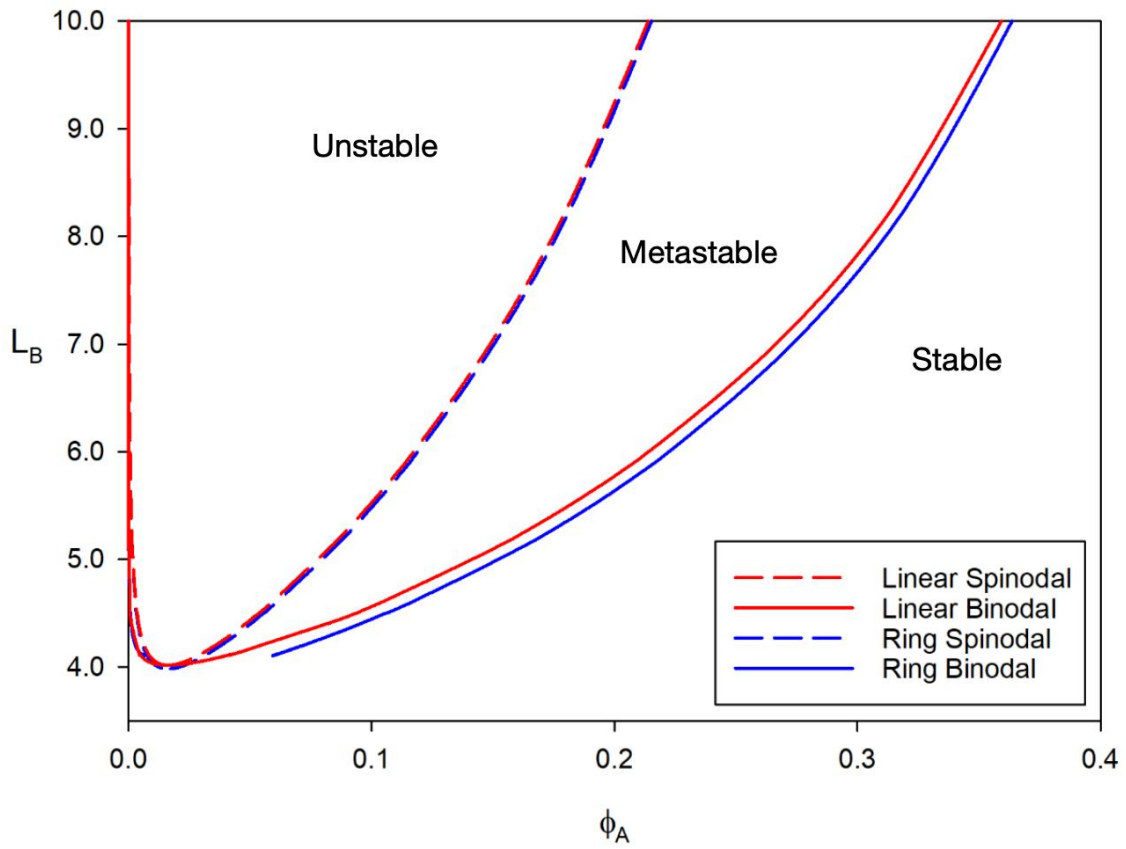


Figure 3-2. Bulk thermodynamic phase diagrams of the L and R systems, showing the binodal (coexistence) and spinodal (stability-limit) boundaries.

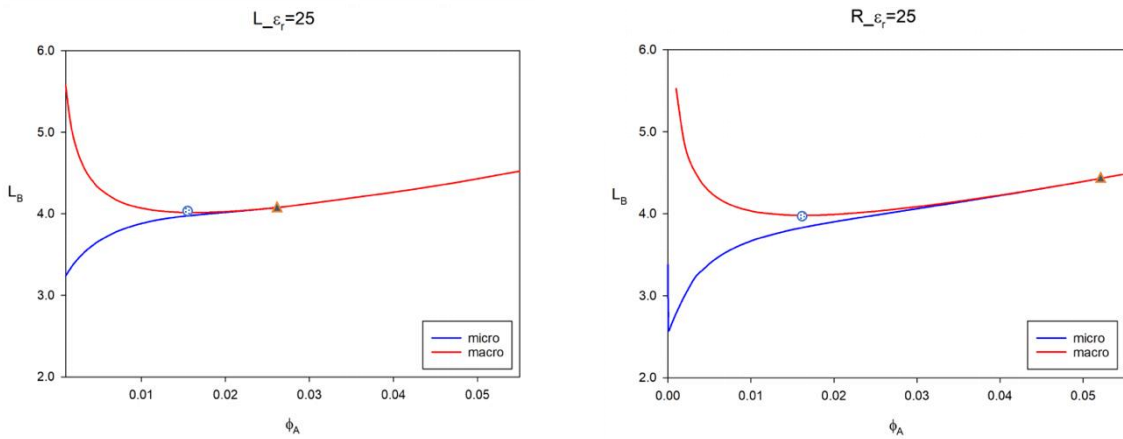


Figure 3-3. RPA analyses of the L (left) and R (right) systems at $\epsilon_r = 25$. Circles denote the macrophase critical points, and triangles mark the onset of microphase separation.

To further investigate the effect of pressure on the shielding between charges, Figure 3-4 presents the bulk thermodynamic and RPA analyses for the L system without vdW attractions at $P = 1kT/\nu^*$ and $P = 5kT/\nu^*$. Increasing the pressure, the macrophase CP shifts toward the PA-poor region, and the microphase region narrows accordingly. Notably, in the bulk thermodynamic phase diagram, a crossover of the two CPs occurs at different pressures, indicating an anomaly that is required to be corrected by considering microphase separation. Figure 3-5 illustrates the phase diagrams under the same pressure conditions when short-range vdW attractions are included. SCFT calculations predict the formation of LAM, HEX, and BCC morphologies. Compared with Figure 3-4, the critical composition ϕ_A decreases, indicating that short-range attractions promote demixing at lower polymer fractions.

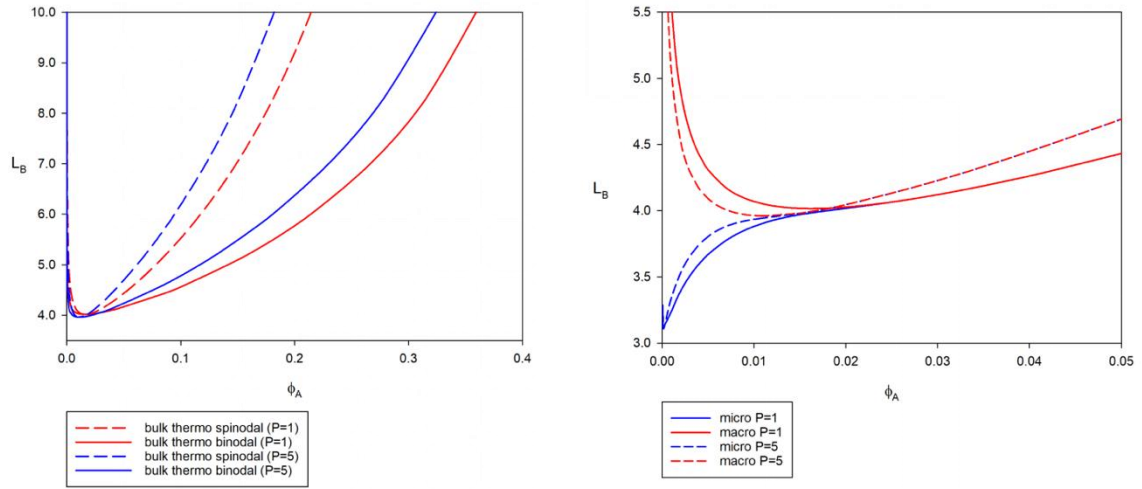


Figure 3-4. Bulk thermodynamic (left) and RPA (right) analyses of the L system without vdW attractions at pressures of $P\nu^*/kT = 1$ and $P\nu^*/kT = 5$.

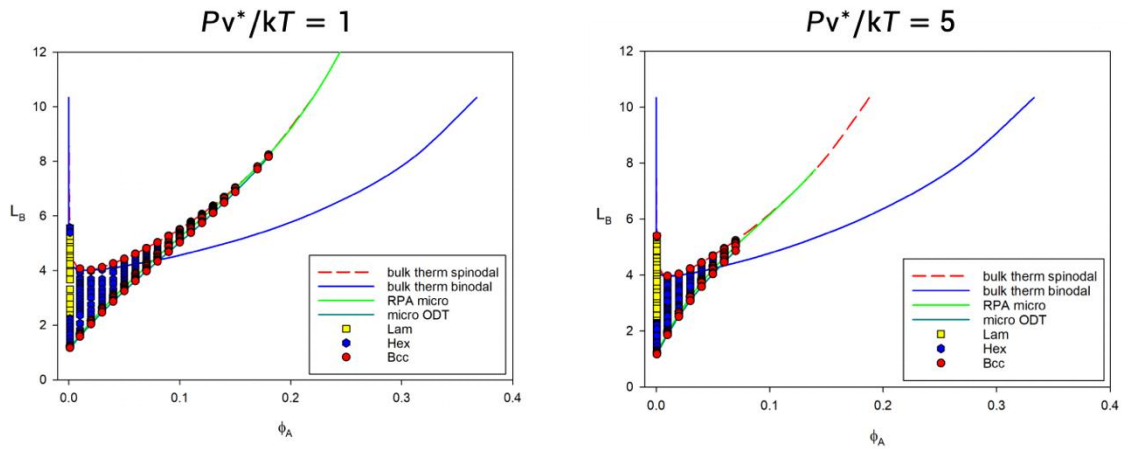


Figure 3-5. Phase diagrams of the L system with vdW attractions at pressures of $P=1$ (left) and 5 (right). Lam, Hex, and Bcc morphologies were obtained from SCFT calculations.

3.3.2 Weakly charged system

To investigate the phase behavior of weakly charged PA/PC blends in salt-free conditions, considered two representative systems—a barotropic and a baroplastic system. The former takes $\bar{\varepsilon}_{AA}/k_B = \bar{\varepsilon}_{CC}/k_B = 4107K$ and $\bar{\varepsilon}_{AC}/\sqrt{\bar{\varepsilon}_{AA}\bar{\varepsilon}_{CC}} = 0.98000$, corresponding to symmetric self-dispersion interactions. The latter takes $\bar{\varepsilon}_{AA}/k_B = 4107K$, $\bar{\varepsilon}_{CC}/k_B = 3000K$, and $\bar{\varepsilon}_{AC}/\sqrt{\bar{\varepsilon}_{AA}\bar{\varepsilon}_{CC}} = 1.00144$, exhibiting asymmetric self-dispersion interactions. The average charge fraction f_o , or equivalently the charge fraction f_A of PA, was fixed to represent a weakly charged condition. The sequence of phase transitions was determined by comparing the free energies of the relevant mesophases. The system undergoes an ODT from the disordered to the BCC phase, followed by spinodal and subsequent OOT from BCC to HEX and then from HEX to LAM upon cooling, analogous to the transition sequence observed in diblock copolymer melts.¹³² For the barotropic system, Figure 3-6 depicts the full phase diagram for $N_A = N_C = 50$ and $f_o = 0.02$, while Figure 3-7 presents the full phase diagram for $N_A = N_C = 50$ and $f_A = 0.02$. Due to global electroneutrality, the charge fraction of the PC chains is given by $f_C = \phi_A f_A / (1 - \phi_A)$, which increases with ϕ_A . All CPs occur at $\phi_A^{CP} = 0.5$, since $f_C = f_A = f_o$ at $\phi_A = 0.5$. Consequently, differences in charge distribution result in a skewed phase diagram in Figure 3-7. Another factor considered was the effect of chain-size asymmetry on the phase behavior. Figure 3-8 presents the full phase diagram for $N_A = 500$, $N_C = 50$, and $f_o = 0.006$. The results indicate that the chain-size asymmetry induces an asymmetric phase diagram, with the CP shifted to $\phi_A^{CP} = 0.41183$.

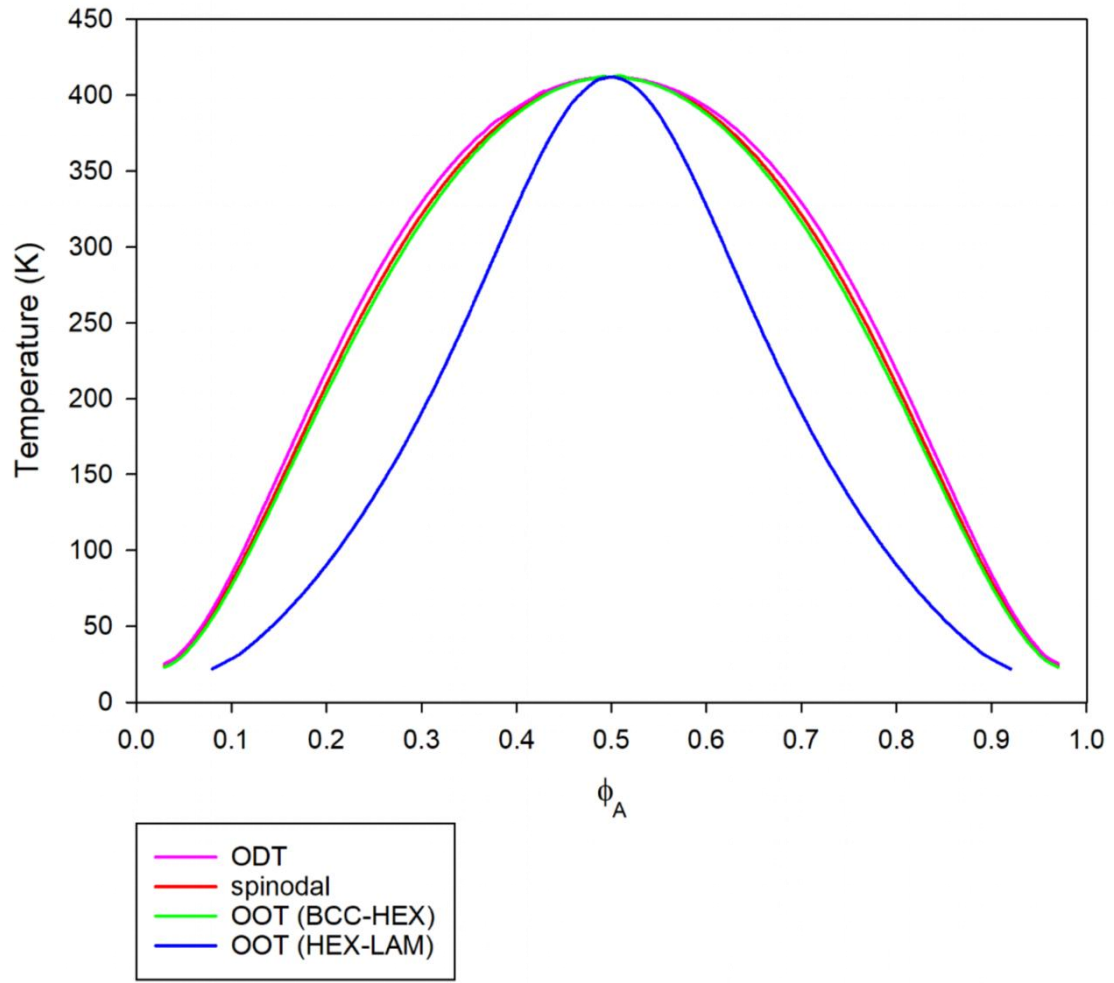


Figure 3-6. Phase diagram for the PA/PC blend with $N_A = N_C = 50$, $f_o = 0.02$, and $\bar{\varepsilon}_{AA} = \bar{\varepsilon}_{CC}$. Temperature for various transitions is plotted against ϕ_A .

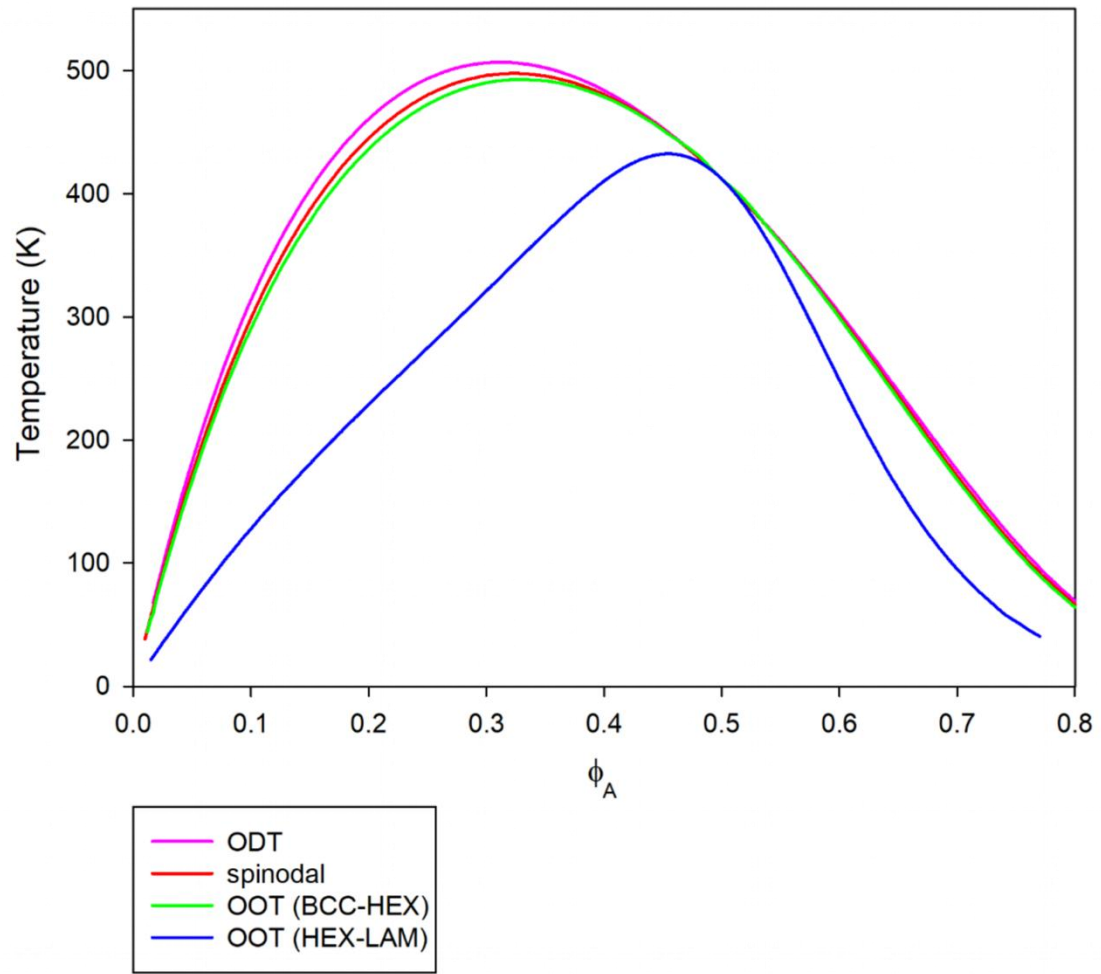


Figure 3-7. Phase diagram for the PA/PC blend with $N_A = N_C = 50$, $f_A = 0.02$, and $\bar{\varepsilon}_{AA} = \bar{\varepsilon}_{CC}$. Temperature for various transitions is plotted against ϕ_A .

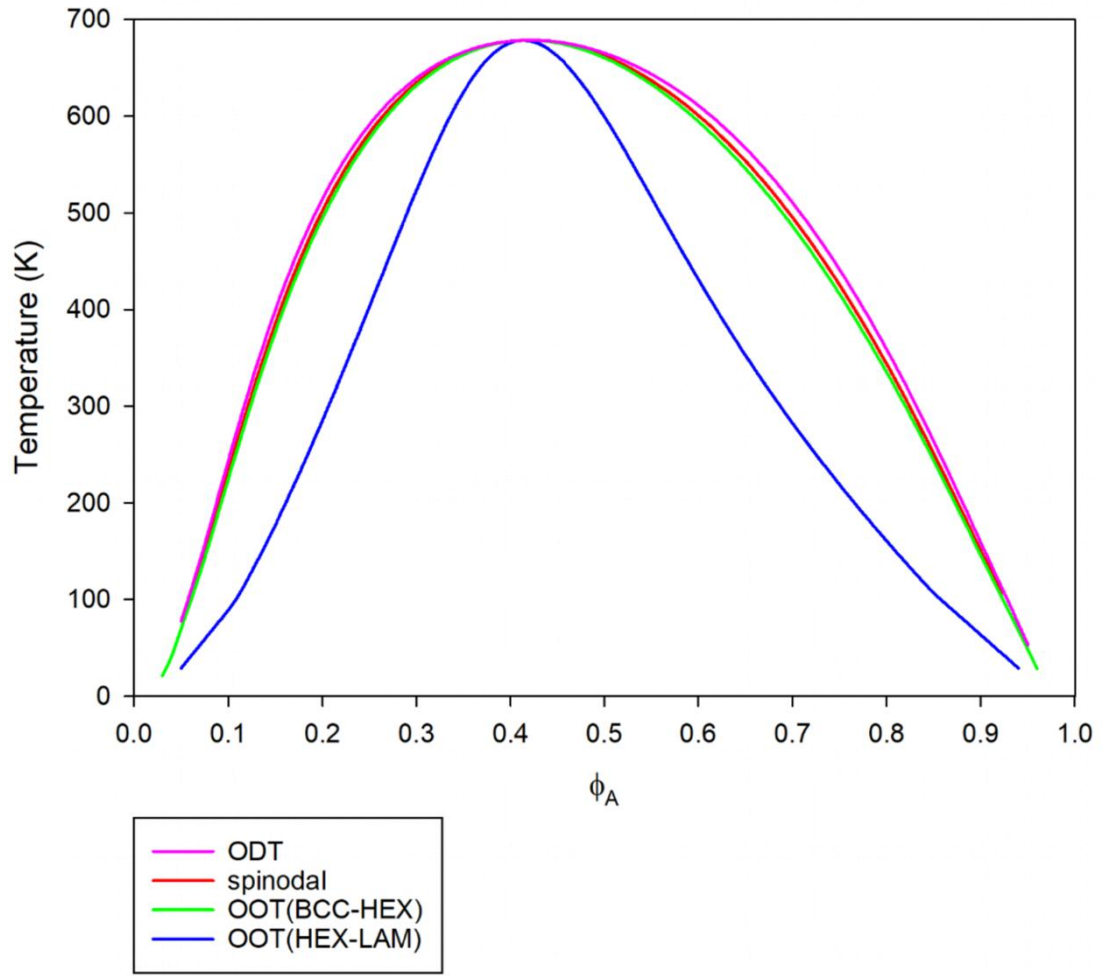


Figure 3-8. Phase diagram for the PA/PC blend with $N_A = 500$, $N_C = 50$, $f_o = 0.006$, and $\bar{\varepsilon}_{AA} = \bar{\varepsilon}_{CC}$. Temperature for various transitions is plotted against ϕ_A .

For the baroplastic system, Figure 3-9 depicts the full phase diagram for $N_A = N_C = 50$ and $f_o = 0.02$, exhibiting a complex hourglass-type phase behavior with the CP appearing in the lower-temperature region at $\phi_A^{CP} = 0.52267$. The skewed phase diagram arises from the disparity in dispersion interactions between the constituents. Figure 3-10 depicts the full phase diagram for $N_A = 100$, $N_C = 50$ and $f_o = 0.02$, exhibiting both asymmetric self-dispersion interactions and chain-size disparity. Compared with Figure 3-9, increasing N_A enhances microphase separation. Notably, due to the chain-size asymmetry, CP occurs at the bottom region ($\phi_A^{CP} = 0.51468$) in addition to the top region ($\phi_A^{CP} = 0.54674$). The higher ϕ_A of the top CP can be attributed to the enhanced volume difference between PA and PC at higher temperatures.

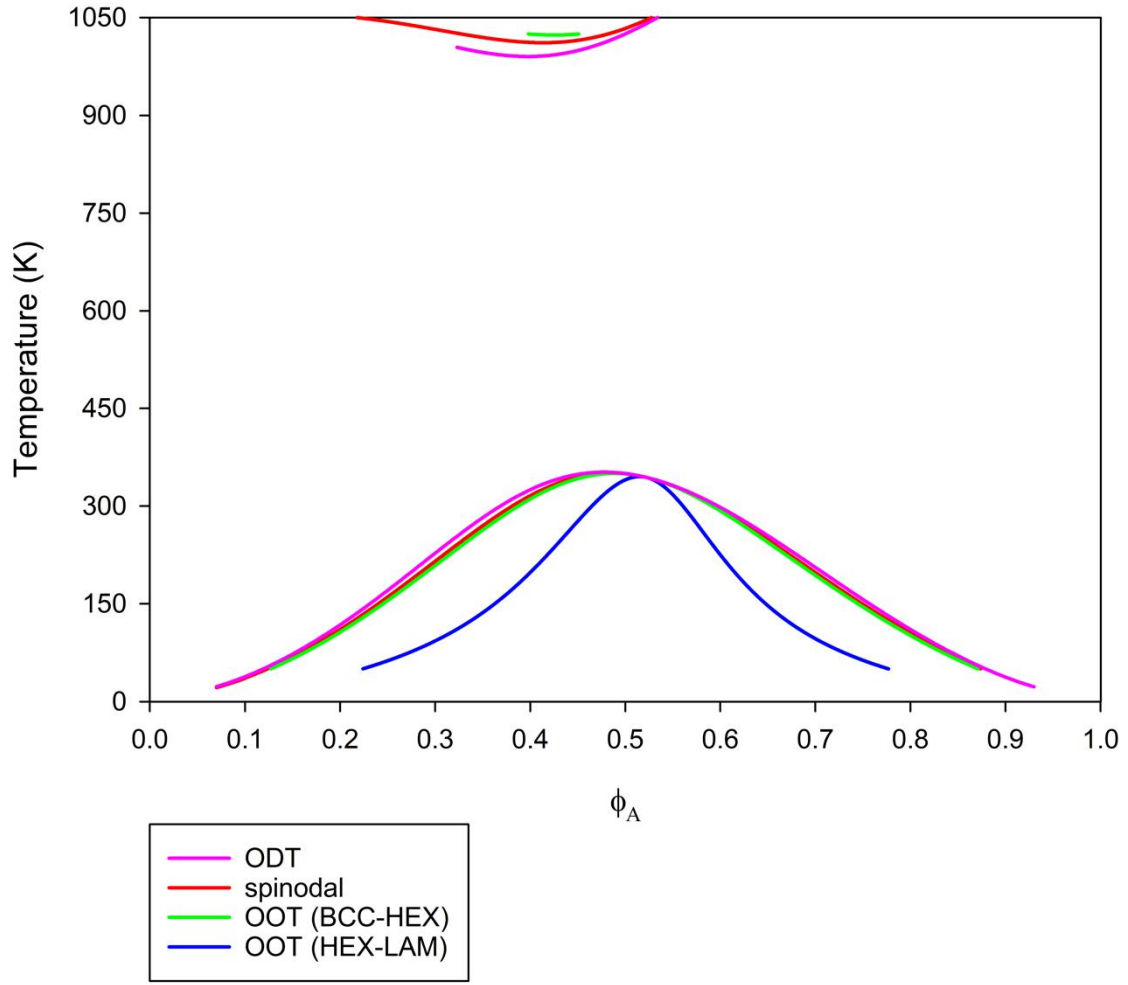


Figure 3-9. Phase diagram for the PA/PC blend with $N_A = N_C = 50$, $f_o = 0.02$, and $\bar{\varepsilon}_{AA} > \bar{\varepsilon}_{CC}$. Temperature for various transitions is plotted against ϕ_A .

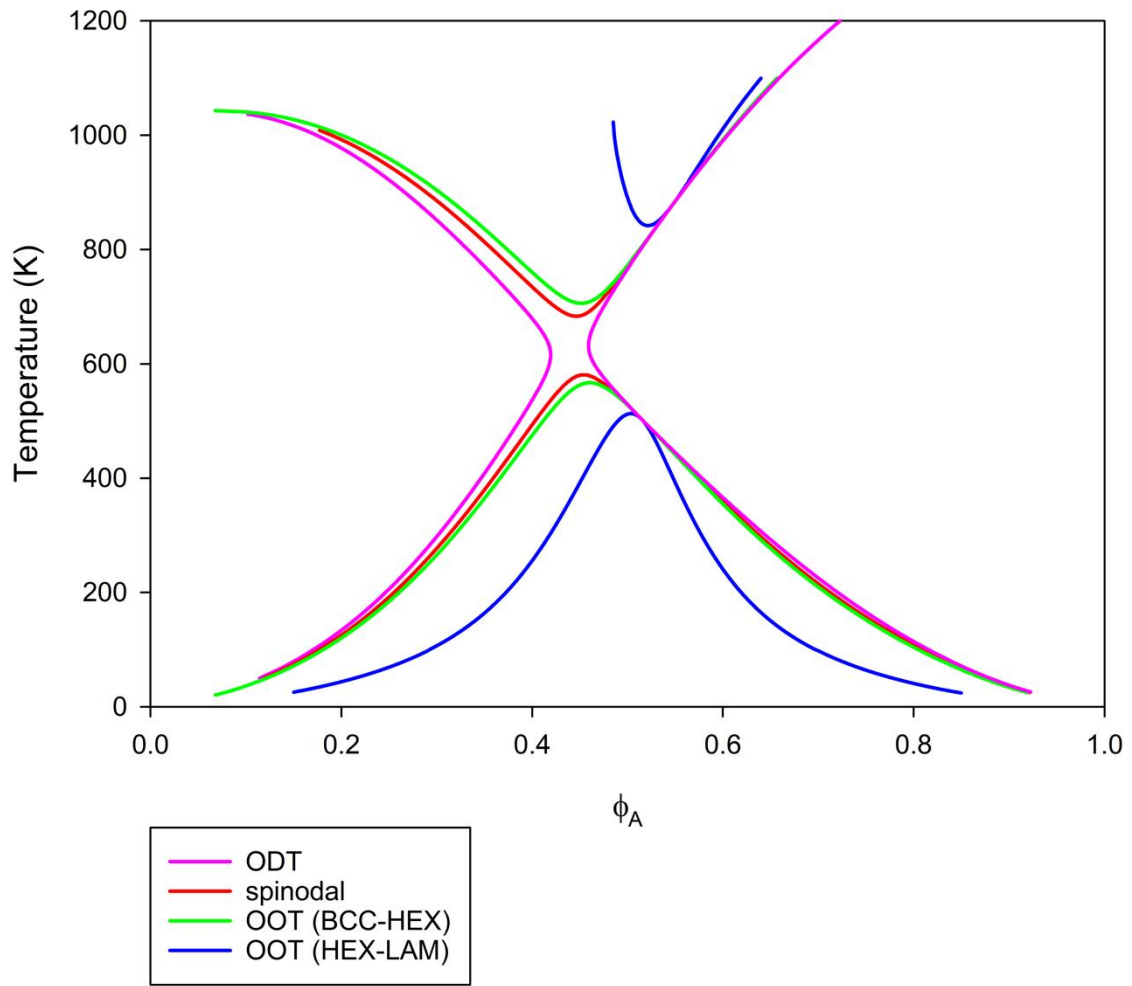


Figure 3-10. Phase diagram for the PA/PC blend with $N_A = 100$, $N_C = 50$, $f_o = 0.02$, and $\bar{\varepsilon}_{AA} > \bar{\varepsilon}_{CC}$. Temperature for various transitions is plotted against ϕ_A .

Finally, we investigated the pressure response of various phase transitions. Figure 3-11 (a) and (b) show the results for the barotropic and baroplastic systems, respectively, both at $N_A = N_C = 50$, $f_o = 0.02$ and $\phi_A = 0.45$. In the barotropic system Figure 3-11 (a), the transition behavior increases monotonically with pressure, whereas in the baroplastic system Figure 3-11 (b), it is suppressed by pressure. This contrasting behavior reflects the competing effects of enthalpic χ_H and entropic χ_S .

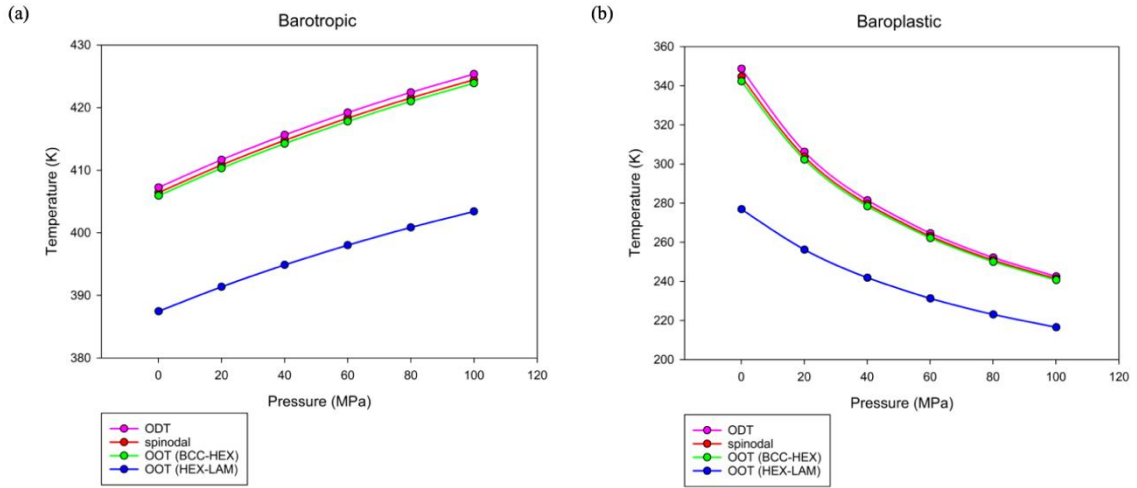


Figure 3-11. Pressure dependence of various ordering transitions for the PA/PC blend ($N_A = N_C = 50$, $f_o = 0.02$, and $\phi_A = 0.45$) in the (a) barotropic and (b) baroplastic systems, respectively.

3.4 Summary

This study systematically investigates the complex phase behavior of polyelectrolyte solutions by constructing a Landau free energy based on RPA theory of Gaussian HSC, combined with a compressible Cho–Sanchez EOS. The sequence of phase transitions was identified from a systematic comparison of the free energy landscapes associated with intermediate phases, analogous to transitions in diblock copolymer melts. The model accounts for excluded volume, chain connectivity, dispersive interactions, and long-range Coulomb interactions. The results reveal that electrostatic shielding induces microphase separation into a PA-rich phase and an almost pure solvent phase, rendering the macroscopic CP aphysical. Notably, the shielding effect appears stronger in the R system than in the L system. For the R system, the macroscopic CP shifts toward the PA-rich region, and the microphase region is broader compared to the L system. Transitioning from fully charged to weakly charged systems, blends with symmetric chain-size and dispersive interactions exhibit a fully symmetric phase diagram with $\phi_A^{CP} = 0.5$, whereas differences in charge distribution induce a skewed phase diagram. By tuning the chain-size asymmetry and dispersion interactions, the PA/PC blends exhibit more complex phase behaviors, including hourglass-type phase diagrams and the emergence of double CPs. Furthermore, the observed barotropic and baroplastic responses open new opportunities for tailoring the phase behavior of weakly charged polyelectrolyte blends in materials design.

IV. Phase Behaviors of Associating A-b-B Diblock Copolymers

4.1 Introduction

Block copolymers, comprised of covalently bonded distinct polymer blocks, have garnered significant and sustained interest due to their ability to self-assemble into ordered nanostructures, establishing them as pivotal materials for applications such as thermoplastic elastomers, drug delivery, and nanolithography.^{64,65} Classical nanostructures include LAM, HEX, and BCC, while more complex morphologies such as bicontinuous mesophase, and Frank–Kasper phases have also been reported.^{50,65,134–136} The classical theoretical framework, based on an incompressible Gaussian chain model, predicts that the phase behavior of diblock copolymers is primarily determined by the degree of polymerization (N), the block volume fraction (ϕ), and the effective Flory interaction parameter (χ).⁶⁰ However, in realistic systems, the A and B blocks of copolymers are not perfectly symmetric, which implies that the incompressibility assumption and monolithic χ parameter are insufficient for capturing complex phase behaviors observed experimentally, particularly the underlying critical transition mechanisms. To address this limitation, this study extends the theoretical framework of polyelectrolyte blend systems to associated AB diblock copolymers. Building on the compressible Landau approach previously developed by Cho, the χ parameter was decomposed into χ_H and χ_S , explicitly incorporating the disparity in self-dispersion interactions between the two blocks, which governs the effective cubic term in the Landau free energy form.^{77–79,86} The Landau free energy was reformulated to provide a unified description of temperature- and pressure-dependent phase behaviors, derived via two mathematically equivalent approaches: one in a direct way using two order parameters, and the other through a transformation of these order parameters.

Focusing on pressure responsiveness in associating A-b-B diblock copolymers, in barotropicity where the ordered phase region expands upon pressurization, polystyrene-b-polybutadiene (PS-b-PBD) exhibits an upper order-disorder transition (UODT) corresponding to UCST behavior,¹³⁷ whereas polystyrene-b-poly(vinyl methyl ether) (PS-b-PVME) exhibits a lower disorder-order transition (LDOT) characteristic of LCST behavior driven by polar bonds C–H $\cdots$ O such as hydrogen bonding.¹³⁸ In baroplasticity, where the ordered phase region shrinks upon pressurization, previously reported polystyrene-b-poly(ethyl hexyl acrylate) (PS-b-PEHA) exhibits a UODT,⁴⁹ polystyrene-b-poly(n-butyl methacrylate) (PS-b-PnBMA) exhibits an LDOT,¹³⁹ and polystyrene-b-poly(n-pentyl methacrylate) (PS-b-PnPeMA) exhibits a unique closed-loop phase behavior.^{37,43} In this study, theoretical phase diagrams were computed using two equivalent formulations of the Landau free energy, and the results were quantitatively compared with experimental observations, thereby confirming the validity and generality of the proposed theoretical framework. Furthermore, the effect of multibody interactions on the phase behavior of associating AB diblock copolymers was also preliminarily examined.

4.2 Theoretical methods

We consider an associating A-b-B diblock copolymer system composed of A and B monomers occupying a total volume (V). Each copolymer chain is modeled as a perturbed tangent HSC, consisting of N_A A-spheres and N_B B-spheres of identical diameter σ . For the k -block, the corresponding close packed volume is given as $V_k = n_c N_k \nu^*$, where $\nu^* = \pi \sigma^3 / 6$ is the monomer volume, the close packed volume fraction is given as $\phi_k = N_k / N_C$, where $N_C = N_A + N_B$ is the copolymer chain size, and the packing density is equal to $\eta_k = \phi_k \eta$, where $\eta = \sum V_k / V$ is the overall packing density. The Helmholtz free energy A per unit volume in the bulk state is expressed as:

$$a = \frac{A}{V} = \frac{A_{id}}{V} + \frac{A_{ni}}{V} = a_{id} + a_{HSC} + a_{pb} + u_{nb-disp} \quad (4-1)$$

where a_{id} is ideal contribution, a_{HSC} is contributions by hard spheres, a_{pb} is contributions by association due to polar bonding, and $u_{nb-disp}$ is dispersion interactions energy between nonbonded monomers. Baxter's adhesive hard sphere potential between g and k -spheres,^{73,74} defined as:

$$\beta u_{gk}(r) = \begin{cases} \infty & r < s_{gk} \\ \ln \left[\frac{12\tau_{gk} (\sigma_{gk} - s_{gk})}{\sigma_{gk}} \right] & s_{gk} < r < \sigma_{gk} \\ 0 & r > \sigma_{gk} \end{cases} \quad (4-2)$$

where $\beta = 1/k_B T$ as usual, and τ_{gk} is temperature related parameter for gk -interactions. The total correlation function h_{gk} is given as:

$$h_{gk}(r) = g_{gk}(r) - 1 = \begin{cases} -1 + \frac{t_{gk}}{\sigma_{gk}} \delta(r - \sigma_{gk}) & r < \sigma_{gk} \\ 0 & r > \sigma_{gk} \end{cases} \quad (4-3)$$

where $g_{gk}(r)$ is the gk -pair distribution function, t_{gk} carries the concept of gk -association probability. Mayer f-function obtained as:

$$f_{gk}(r) = e^{-\beta u_{gk}(r)} - 1 = -1 + \frac{\sigma_{gk}}{12\tau_{gk}} \delta(r - \sigma_{gk}) \quad r < \sigma_{gk} \quad (4-4)$$

The cavity function y_{gk} is given as

$$y_{gk} = e^{\beta u_{gk}(r)} g_{gk}(r) = g_{gk}(r) - c_{gk}(r) \quad (4-5)$$

where $c_{gk}(r)$ is the direct correlation function. Its contact value is given by

$$y_{gk}(\sigma) = \frac{h_{gk}(r) + 1}{f_{gk}(r) + 1} = \frac{12\tau_{gk}t_{gk}}{\sigma_{gk}^2} \quad (4-6)$$

The t_{gk} values of the cavity function require

$$\pi \sum_i \rho_i \left[2t_{gi}t_{ik} - \frac{1}{\Delta} t_{ig}\sigma_i\sigma_k - \frac{1}{\Delta} t_{ik}\sigma_i\sigma_g + \frac{\sigma_g\sigma_k\sigma_i^2}{4\Delta^2} \right] - \frac{12\tau_{gk}t_{gk}}{\sigma_{gk}} = -\frac{\sigma_{gk}}{\Delta} \quad (4-7)$$

where ρ_i is the number density of i -spheres, and $\Delta = 1 - \eta$, for connected pairs are

given as $4\pi\rho_k t_{gk}\sigma_{gk} = 1$. To simplify the system, assume $\sigma_{gk} = \sigma$; $\tau_{gk} = \tau \equiv$

$a_\tau T$, thus t_{gk} for association simplified as $\frac{t_{gk}}{\sigma^2} = \frac{6\eta}{4\Delta^2} \left[12\tau - 2 + \frac{6\eta}{\Delta} \right]^{-1}$. The

compressibility pressure equation is given by

$$\begin{aligned}
\beta P \nu^* &= \beta P_{HSC} \nu^* - \frac{\eta_A \eta_B}{1-\eta} \left(2 + \frac{3\eta}{1-\eta} \right) \cdot \frac{6t_{gk}}{\sigma^2} \\
&= \beta P_{HSC} \nu^* - \frac{\eta_A \eta_B}{1-\eta} \left(2 + \frac{3\eta}{1-\eta} \right) \cdot \frac{9\eta}{(1-\eta)^2} \left[12\tau - 2 + \frac{6\eta}{1-\eta} \right]^{-1} \\
&= \left[\frac{3(\eta^2 + \eta^3)}{2(1-\eta)^3} + \frac{\eta}{N_C} \frac{(1+\eta/2)}{(1-\eta)^2} \right] - \frac{\eta_A \eta_B}{1-\eta} \left(2 + \frac{3\eta}{1-\eta} \right) \\
&\quad \cdot \frac{9\eta}{(1-\eta)^2} \left[12\tau - 2 + \frac{6\eta}{1-\eta} \right]^{-1}
\end{aligned} \tag{4-8}$$

Therefore, the Helmholtz free energy each contribution is given as follows

$$\beta a_{id} \nu^* = \frac{\eta}{N_C} \ln \frac{\eta K}{N_C} \tag{4-9}$$

$$\beta a_{HSC} \nu^* = \frac{3}{2} \left[\frac{\eta}{(1-\eta)^2} - \left(\eta - \frac{\eta}{N_C} \right) \frac{1}{1-\eta} \right] - \frac{\eta}{N_C} \left[\ln(1-\eta) + \frac{3}{2} \right] \tag{4-10}$$

$$\begin{aligned}
&\beta a_{pb} \nu^* \\
&= \\
&- \frac{3\eta_A \eta_B}{4\eta} \left[3 \left(\frac{1}{(1-\eta)^2} - 1 \right) - 12\tau \left(\frac{1}{1-\eta} - 1 \right) \right. \\
&- 2(1-8\tau) \\
&\quad \left. + 12\tau^2 \ln \left(\frac{(1-\eta)(1-6\tau)}{1-4\eta-6\tau(1-\eta)} \right) \right]
\end{aligned} \tag{4-11}$$

$$\beta u_{nb-disp} \nu^* = \frac{1}{2} \beta \sum_{gk} \phi_g \phi_k \varepsilon_{gk} u(\eta) \eta = \frac{1}{2} \beta \sum_{gk} \eta_g \eta_k \varepsilon_{gk} \frac{u(\eta)}{\eta} \tag{4-12}$$

where K is a molecular constant that does not affect any thermodynamic calculations, ε_{gk} is the characteristic gk -sphere contact energy. The effective RPA interaction field is given by the excess chemical potential:

$$W_{gk} = \frac{\partial^2 a_{ni} \nu^*}{\partial \eta_g \partial \eta_k} \bigg|_{T,V,ns} = \frac{\partial \left[\mu_k^{ex} / N_k \right]}{\partial \eta_g} \bigg|_{T,V,ns} \quad (4-13)$$

In our A-b-B diblock copolymer system, structure factor from quadratic term, consisting of Guassian part and local interactions are given as:

$$[S_{gk}^{-1}] = [S_{gk}^{0-1}] + [\beta D_{gk} a_{ni}] \quad (4-14)$$

where $D_{gk} a_{ni} \equiv \partial^2 a_{ni} / \partial \eta_g \partial \eta_k$ to give the effective local interactions in two-body level. The spinodals, representing the stability boundaries, are defined by the condition $\frac{\Gamma_{AA}}{\Gamma_{AB}} = \frac{\Gamma_{AB}}{\Gamma_{BB}}$, which can be expressed in the Landau free energy as:

$$\bar{\Gamma}_{11} - \frac{\bar{\Gamma}_{12}^2}{\bar{\Gamma}_{22}} = 0 \quad (4-15)$$

The effective Florey parameter χ is very important for the melt phase behavior of diblock copolymers. It consists of two contributions as $\chi = \chi_H + \chi_S$.^{79,140} The former χ_H is given by the non-Gaussian part of $\bar{\Gamma}_{11}$:

$$\chi_H = \eta (D_{AA} - 2D_{AB} + D_{BB}) (-\beta u_{ub-disp}) / 2 = \frac{1}{2} \beta \Delta \bar{\varepsilon} |u(\eta)| \quad (4-16)$$

where $\Delta \bar{\varepsilon} = \bar{\varepsilon}_{AA} + \bar{\varepsilon}_{BB} - 2\bar{\varepsilon}_{AB}$ implies the exchange energy.

Since $u(\eta)$, χ_H possesses density dependence. $\bar{\Gamma}_{12}$ is given as

$$\bar{\Gamma}_{12} = \frac{\eta}{2} \beta \nu * (D_{AA} a_{ni} - D_{BB} a_{ni}) = \frac{1}{2} \beta (\bar{\varepsilon}_{AA} - \bar{\varepsilon}_{BB}) \cdot \eta \frac{du}{d\eta} \quad (4-17)$$

where $\bar{\varepsilon}_{AA} - \bar{\varepsilon}_{BB}$ represents indicates disparity in self-dispersion interactions between A and B blocks. The latter χ_S represents the entropic contribution to phase stability as

$$\chi_S = \eta (D_{AA} - 2D_{AB} + D_{BB}) (-\beta a_{pb}) / 2 + \frac{1}{2\eta} \cdot \frac{\bar{\Gamma}_{12}^2}{\bar{\Gamma}_{22}} \quad (4-18)$$

Upon pressurization, χ_H and χ_S terms respond oppositely behaviors. η increases, leading to an increase in χ_H , whereas χ_S decreases. If $|\bar{\varepsilon}_{AA} - \bar{\varepsilon}_{BB}| \rightarrow 0$, then $\frac{\chi_S}{\chi} \rightarrow 0$, and the χ_H predominates, which is the barotropicity. In contrast, when the difference remains finite or more sizable, $\frac{\chi_S}{\chi}$ becomes significant, while χ_S and χ are suppressed, which is the baroplasticity.

4.3 Results and discussion

4.3.1 Spinodals from RPA and density profiles from SCFT

First considered an associating AB diblock copolymer system with $N_A = 450$, $N_B = 550$, and a total chain length of $N_C = N_A + N_B = 1000$, to demonstrate the capability of our model in describing associative interactions. Figure 4-1 illustrates the effect of the interaction contrast parameter $\xi = \frac{\varepsilon_{AB}}{\sqrt{\varepsilon_{AA}\varepsilon_{BB}}}$ on the loop-forming spinodals obtained from RPA analysis and the ordered-phase density profiles from SCFT. The results indicate that, as ξ approaches 1, the driving force for phase separation is weakened, and the closed-loop region becomes smaller. The copolymer is ordered inside the loop and disordered outside it.

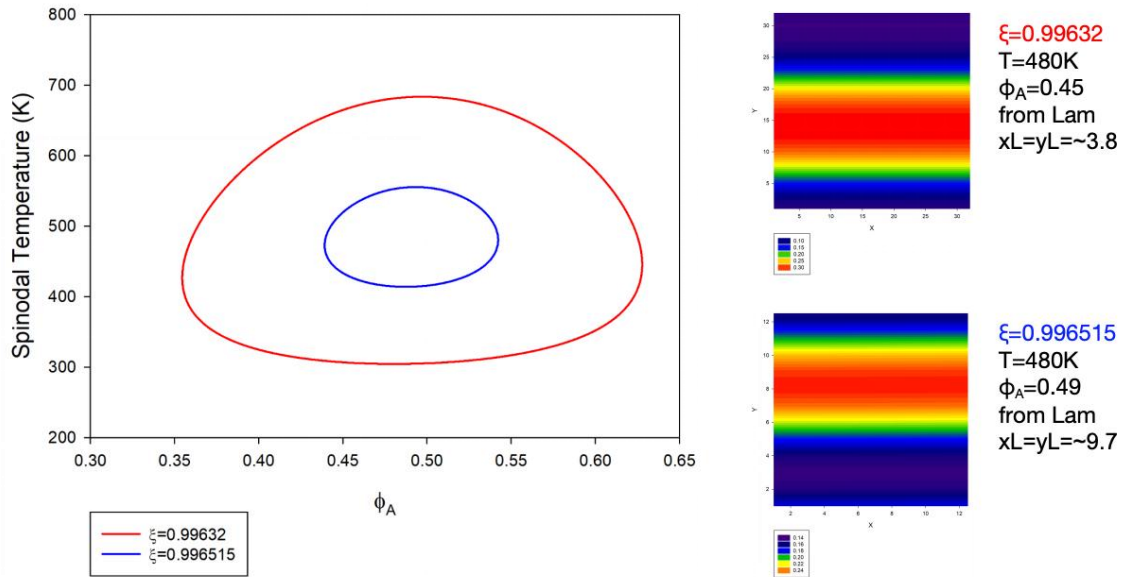


Figure 4-1. Loop-forming spinodals from RPA (left) and density profiles from the ordered phase (Lam) obtained from SCFT (right) at different values of the interaction contrast parameter ξ .

In the baroplastic system, the spinodals of PS-b-PnBMA were computed, and the corresponding density profiles were obtained from SCFT at a pressure of 0.1 MPa, as shown in Figure 4-2. The results reveal an LDOT behavior. Furthermore, the effect of pressure was investigated (Figure 4-3), showing that upon pressurization, the LDOT shifts to higher temperatures with a slope of $\Delta T / \Delta P = 1.965$ K/MPa, closely matching the experimental result ($\Delta T / \Delta P = 1.95$ K/MPa). Figure 4-2 and Figure 4-3 confirm that the proposed model successfully reproduces the experimental phase behavior of PS-b-PnBMA.

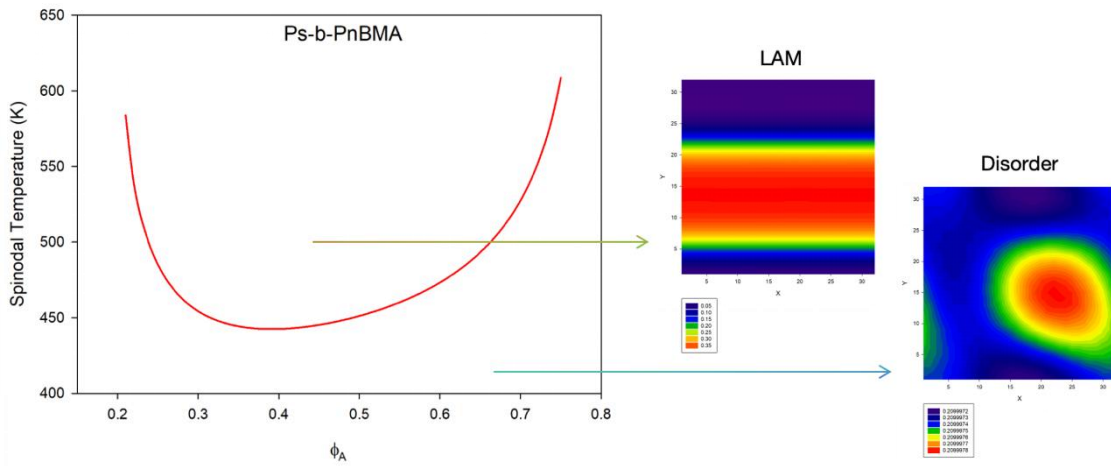


Figure 4-2. Spinodal of PS-b-PnBMA from RPA (left) and density profiles of the ordered (LAM) and disordered phases from SCFT (right) at 0.1 MPa. The disordered phase is identified by the nearly uniform density.

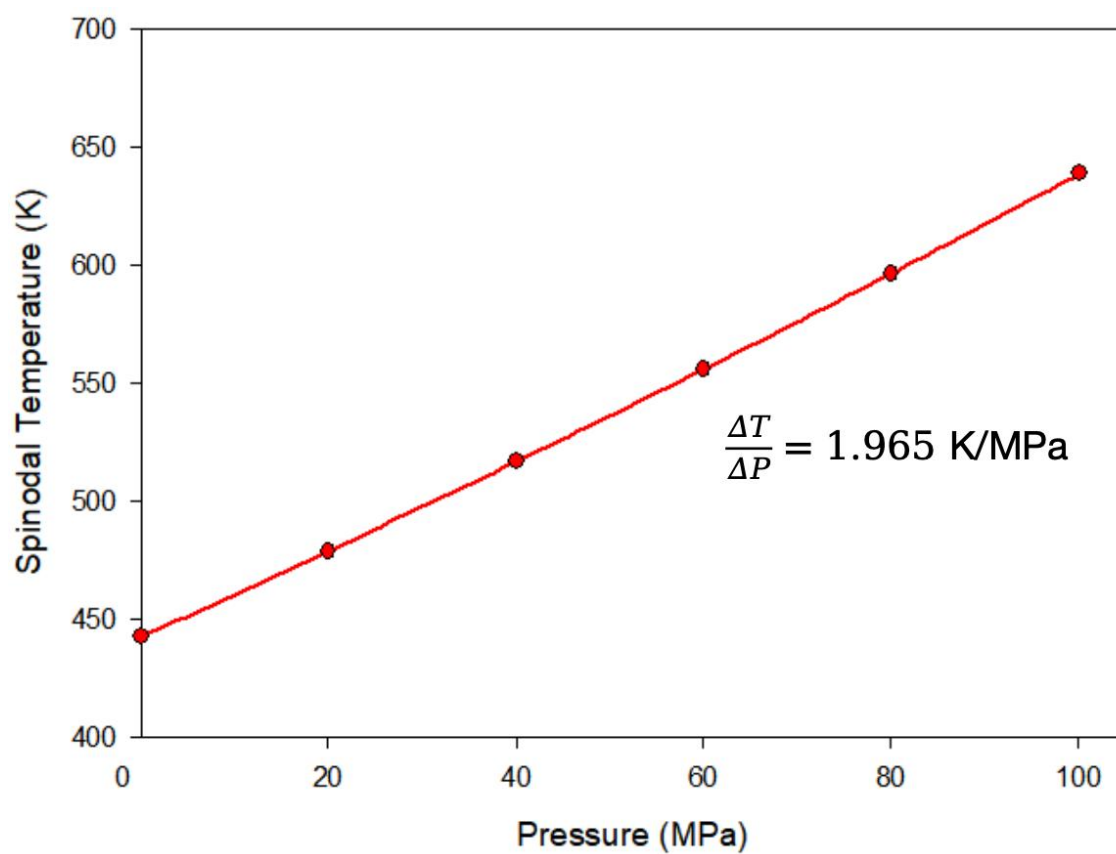


Figure 4-3. Pressure dependence of spinodal temperature of PS-b-PnBMA.

Figure 4-4 presents the spinodal analysis for PS-b-PnPeMA at a pressure of 0.1 MPa, which exhibits a closed-loop phase behavior. The corresponding pressure dependence is shown in Figure 4-5. Pressurization results in the LDOT shifting upward ($\Delta T/\Delta P = 14.26$ K/MPa), whereas the UODT shifts downward ($\Delta T/\Delta P = -14.25$ K/MPa), thereby reducing the ordered-phase region. The results demonstrate that the theoretical predictions for PS-b-PnPeMA are in excellent agreement with the experimentally reported trends. Table 4-1 lists the parameters set used for these two AB diblock copolymers.

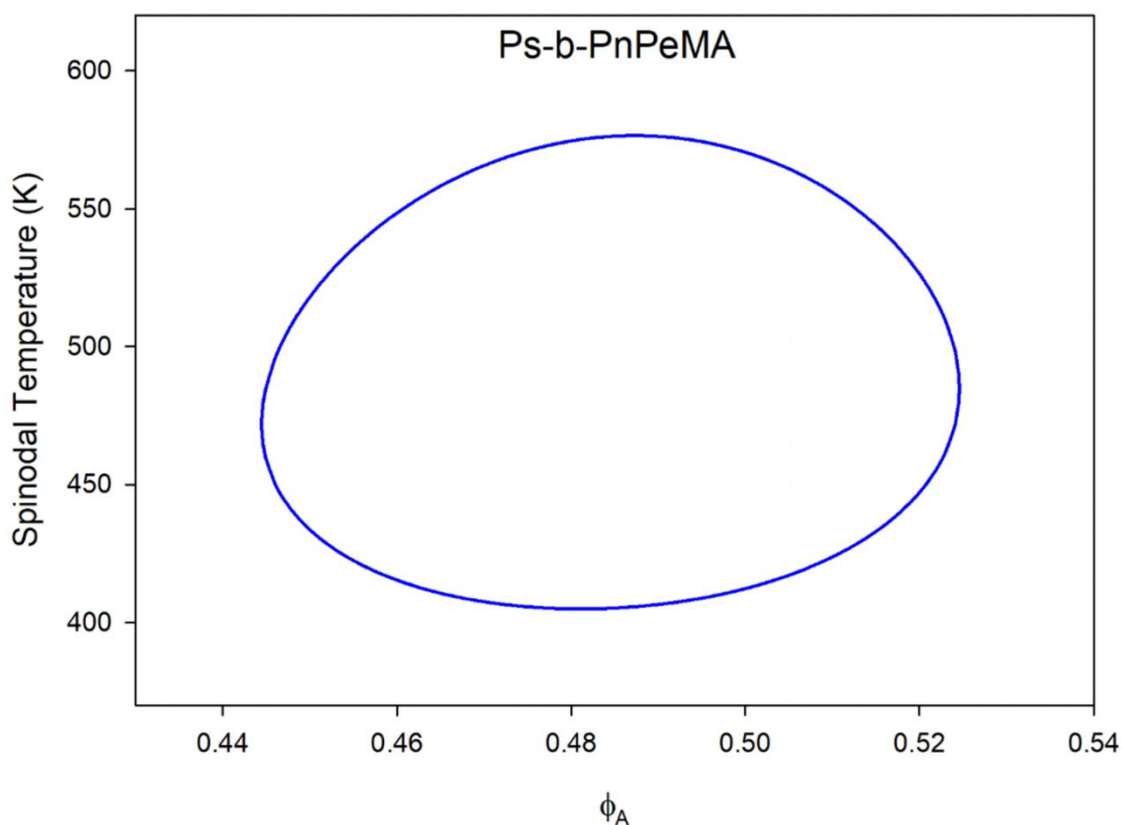


Figure 4-4. Spinodal of PS-b-PnPeMA from RPA at 0.1 MPa.

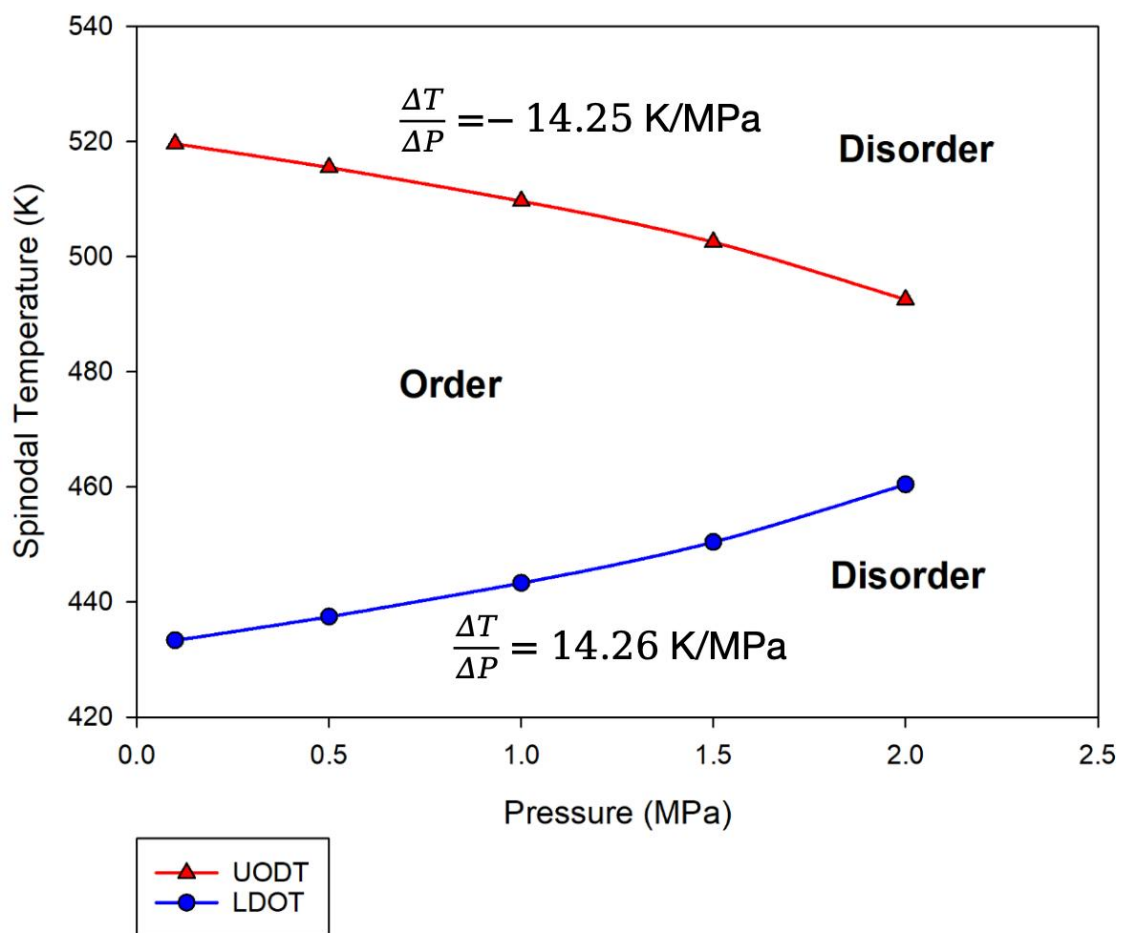


Figure 4-5. Pressure dependence of spinodal temperature of PS-b-PnPeMA.

Table 4-1. Model parameters for the AB diblock copolymers.

Block copolymers	N_A	N_B	a_τ	ξ
PS-b-PnBMA	557.2455	557.6841	0.01468	0.9879
PS-b-PnPeMA	510.7916	510.9412	0.029515	0.9946

4.3.2 Temperature dependence of various phase transitions

The sets of homopolymer parameters are given in Table 4-2. The various phase behaviors of the associating AB diblock copolymers were analyzed by determining the equilibrium mesophases and their stability through a comparison of the Landau free energy minima. Figure 4-6 presents the phase diagram of a typical UODT-type diblock copolymer, PS-b-PBD, with $N_C = 400$ and other parameters used in Table 4-2. Figure 4-6 (a) and (b) illustrate the transition temperature and the relevant parameter for phase segregation ($N_C \chi$), respectively, plotted as functions of ϕ_A . Owing to the small disparity in self-dispersion interactions ($|\bar{\varepsilon}_{PS} - \bar{\varepsilon}_{PBD}|$), the CP appears at $\phi_A = 0.50095$ and $N_C \chi = 10.49494$, resulting in an almost symmetric phase diagram.

Table 4-2. Molecular parameters for the homopolymers of the AB diblock copolymers.

Parameters	PS (A)	PBD (B)	PVME (B)	PEHA (B)	PnPrMA (B)
σ (Å)	4.039	4.039	3.900	3.840	4.000
$\bar{\varepsilon}_{gg}/k$ (K)	4107.0	4065.9	3644.8	3755.7	3835.1
ξ_{AB}	—	0.99565	1.00264	0.99880	1.00090

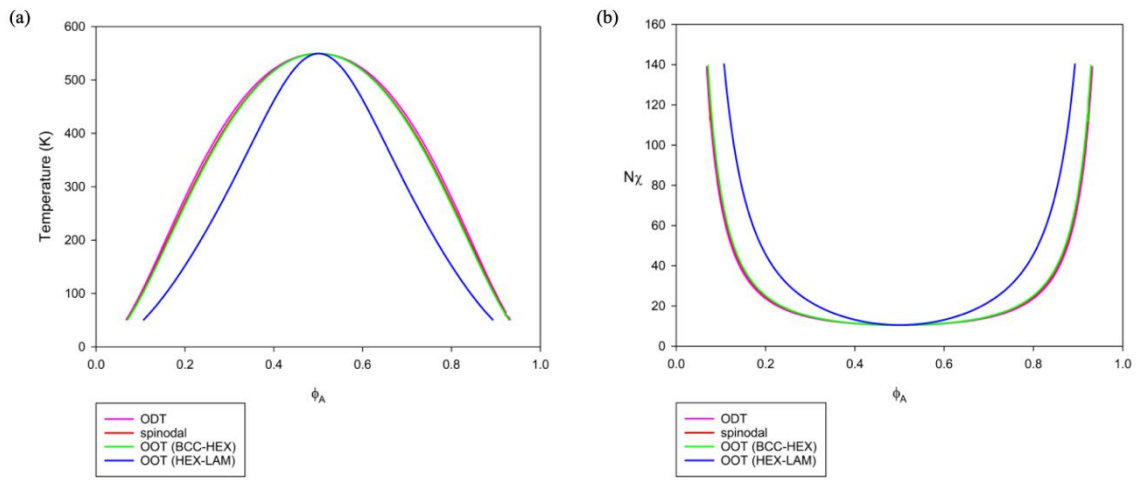


Figure 4-6. Phase diagram of PS-b-PBD with $N_C = 400$ as a function of the PS (A) volume fraction ϕ_A : (a) absolute temperature and (b) the relevant parameter $N_C \chi$.

Figure 4-7 presents the phase diagram of a LDOT-type diblock copolymer, PS-b-PVME, with $N_C = 20000$ and other parameters used in Table 4-2. The spinodal is markedly skewed due to the large disparity in self-dispersion interactions ($|\bar{\varepsilon}_{PS} - \bar{\varepsilon}_{PVME}|$), causing the threshold to shift toward the more compressible PVME-rich domain. However, the CP appears at $\phi_A = 0.50974$, slightly biased toward the less compressible PS-rich domain, which due to $\bar{\varepsilon}_{PS} > \bar{\varepsilon}_{PVME}$. Furthermore, this phenomenon is also observed in the baroplastic LDOT-type polystyrene-b-poly(n-propyl methacrylate) (PS-b-PnPrMA) system with $N_C = 18000$ (Figure 4-8), where the CP is found to be at $\phi_A = 0.50694$.

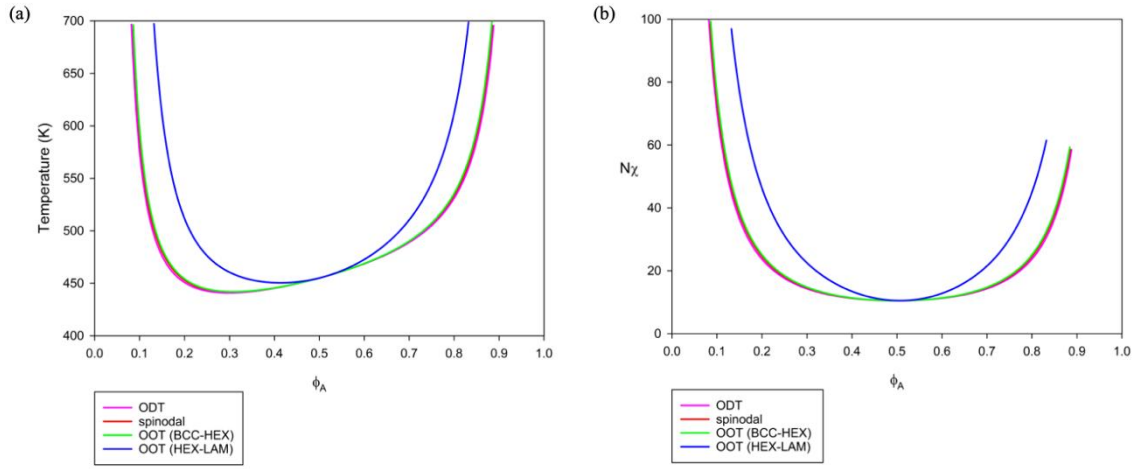


Figure 4-7. Phase diagram of PS-b-PVME with $N_C = 20000$ as a function of the PS (A) volume fraction ϕ_A : (a) absolute temperature and (b) the relevant parameter $N_C \chi$.

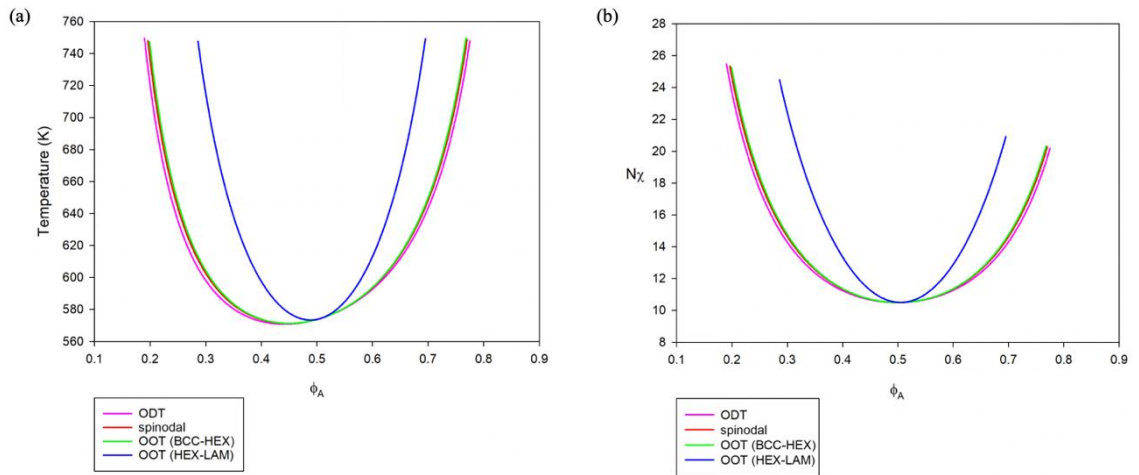


Figure 4-8. Phase diagram of PS-b-PnPrMA with $N_C = 18000$ as a function of the PS (A) volume fraction ϕ_A : (a) absolute temperature and (b) the relevant parameter $N_C \chi$.

4.3.3 Pressure dependence of various phase transitions

Figure 4-9 depicts the pressure dependence of various transitions in the PS-b-PBD system at $\phi_A = 0.4625$. Upon pressurization, the transition temperatures increase, reflecting a barotropic response. In contrast, Figure 4-10 shows the pressure dependence of various transitions in the PS-b-PEHA system with $N_C = 529.5814$ at $\phi_A = 0.440885$, which exhibits a decreasing trend, indicative of a baroplastic response. These theoretical predictions of pressure dependence are in agreement with the experimental observations.

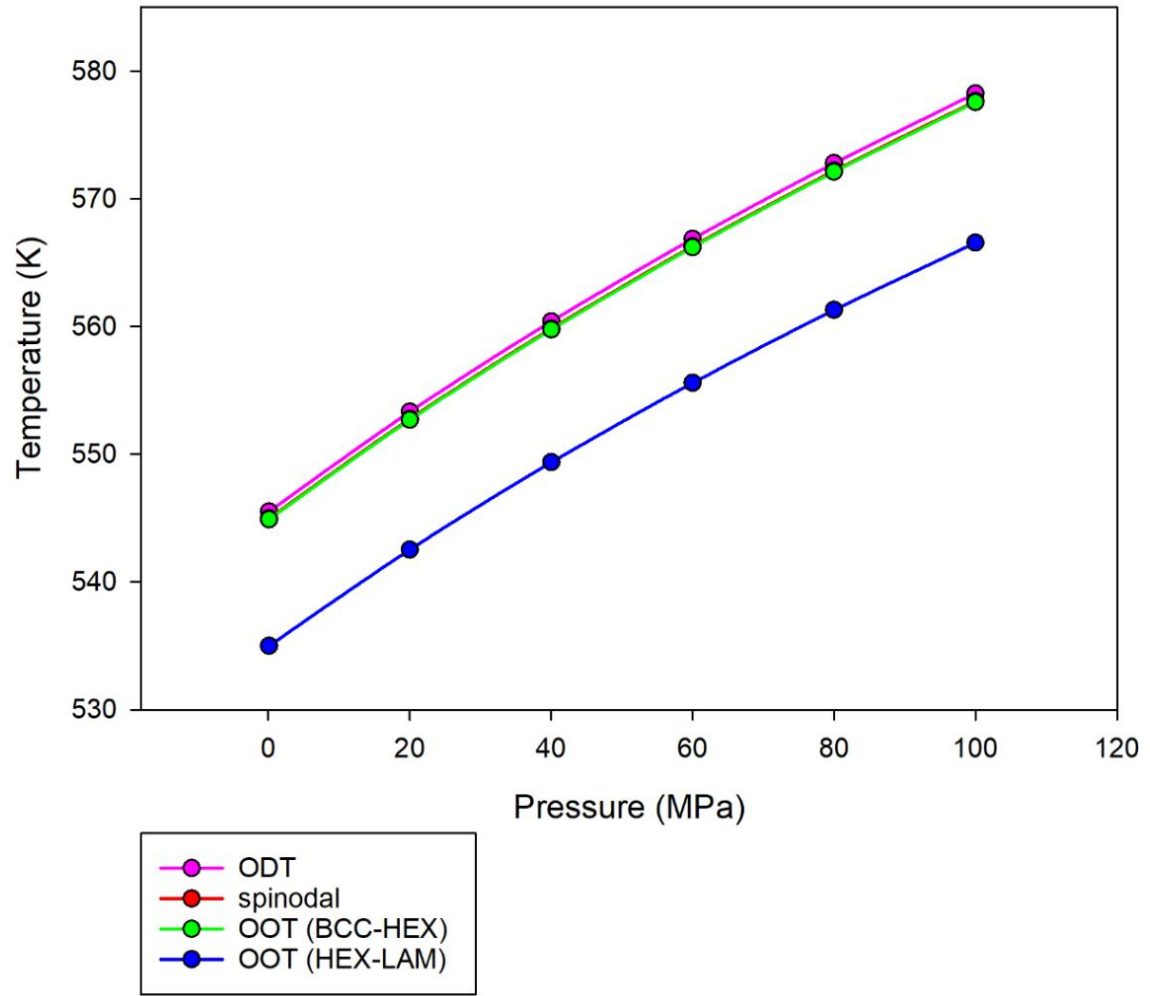


Figure 4-9. Pressure response of various transitions for PS-b-PBD with $N_C = 400$ at $\phi_A = 0.4625$.

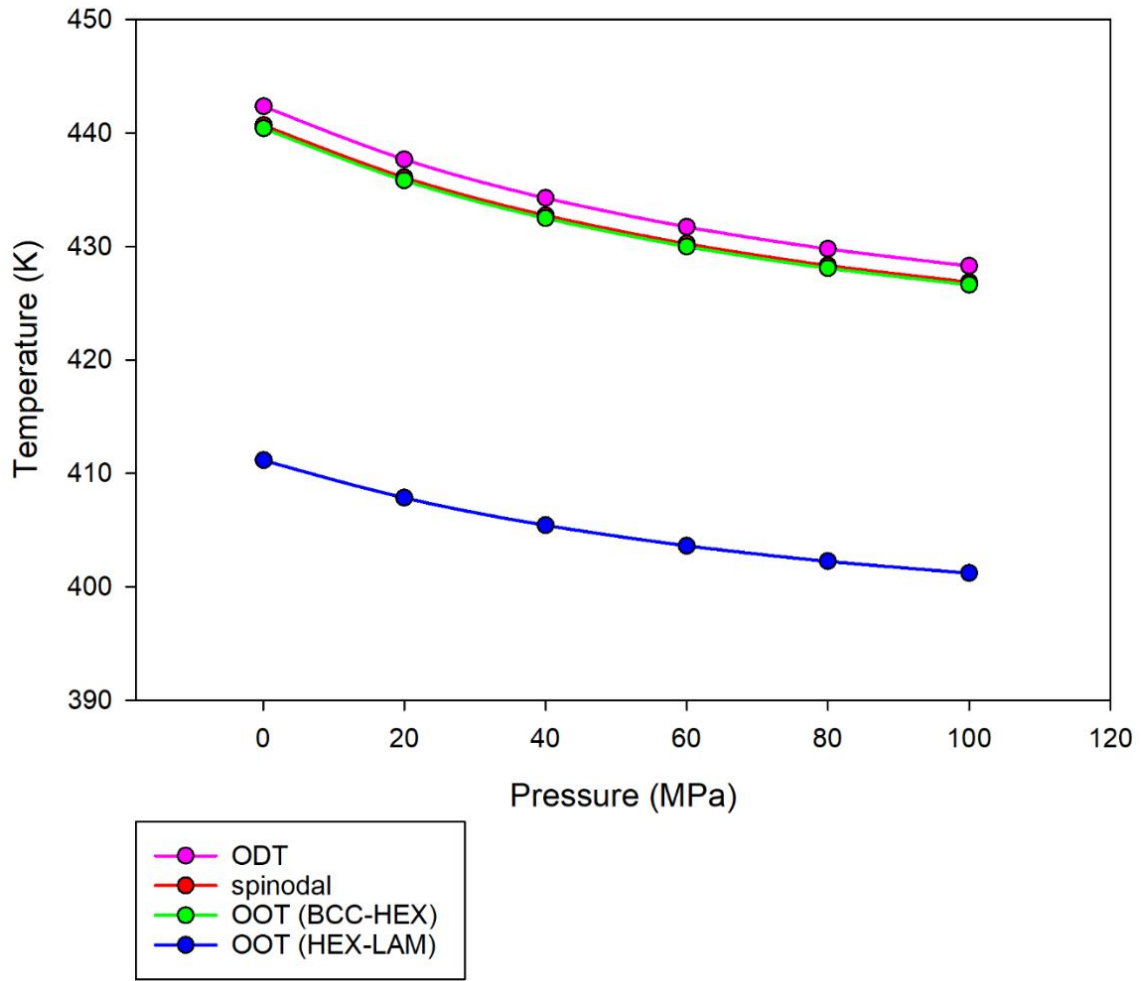


Figure 4-10. Pressure response of various transitions for PS-b-PEHA with $N_C = 529.5814$ at $\phi_A = 0.440885$.

4.3.4 Effect of multibody interactions on phase boundaries

Since mesoscale interactions are inherently density-dependent, it is necessary to adopt a multibody approach to capture the effective interactions in block copolymers accurately. Based on this consideration, we further examined representative UODT- and LDOT-type diblock copolymers by incorporating multibody interactions and comparing the resulting phase diagrams with those at the two-body level. Figure 4-11 presents the phase diagram of the UODT-type PS-b-PBD system obtained by incorporating multibody interactions. Compared with the two-body level result (Figure 4-6), the phase diagram remains essentially unchanged, and the CP remains at $\phi_A = 0.50095$. In contrast, for the LDOT system barotropicity PS-b-PVME, the introduction of multibody interactions leads to a pronounced shift of the CP in Figure 4-12, where $\phi_A = 0.243$. A similar influence on the phase behavior is also observed in the baroplasticity PS-b-PnPrMA system (Figure 4-13), with the CP at $\phi_A = 0.40615$. These results demonstrate that multibody effects strongly influence the enthalpy–entropy balance governing LDOT phase behaviors. Moreover, the distinct discrepancies between the two-body and four-body levels observed in both barotropic and baroplastic systems demonstrate that pressure-dependent compressibility enhances density-dependent interactions in associating AB diblock copolymers.

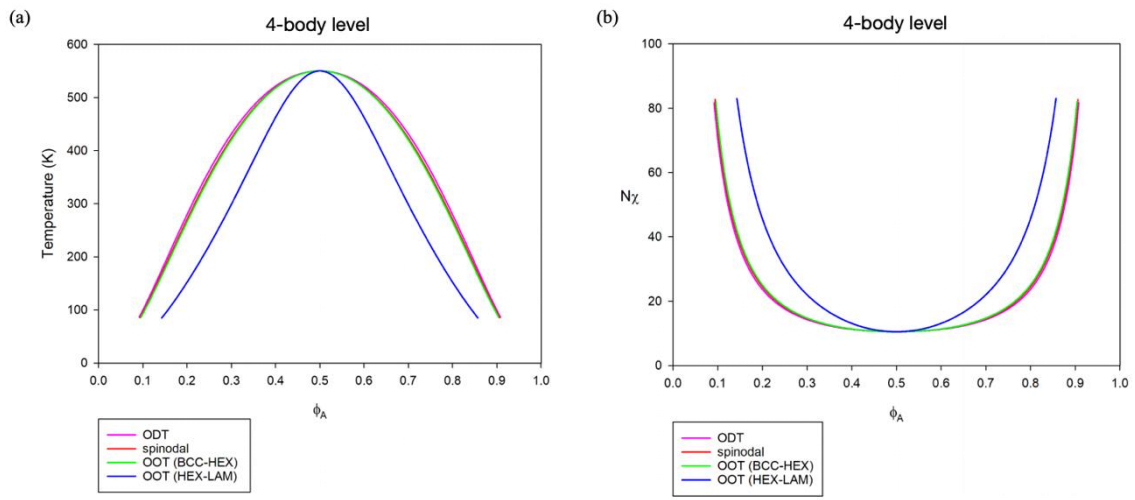


Figure 4-11. Phase diagram of PS-b-PBD with $N_C = 400$ at the four-body level, plotted as a function of ϕ_A : (a) absolute temperature and (b) the relevant parameter $N_C \chi$.

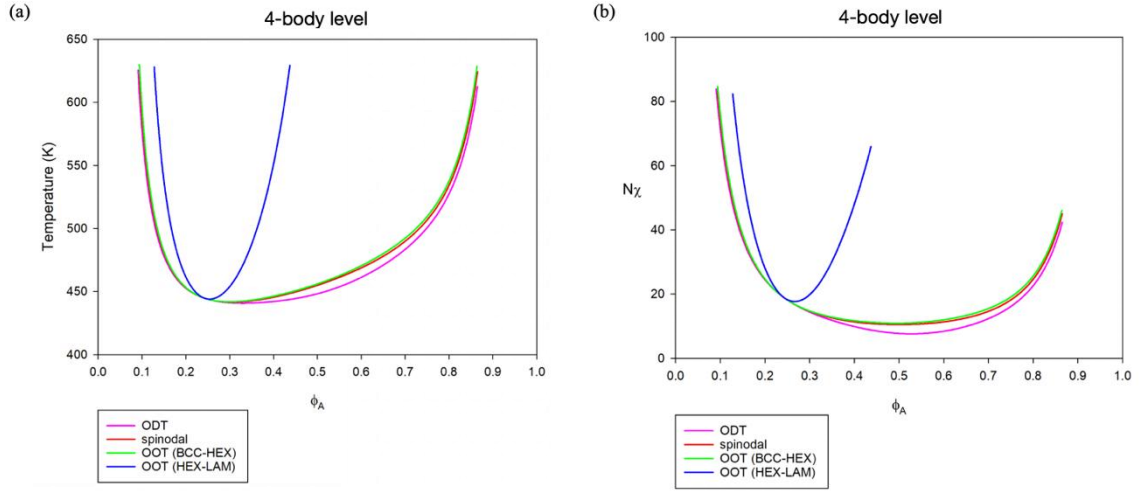


Figure 4-12. Phase diagram of PS-b-PVME with $N_C = 20000$ at the four-body level, plotted as a function of ϕ_A : (a) absolute temperature and (b) the relevant parameter $N_C \chi$.

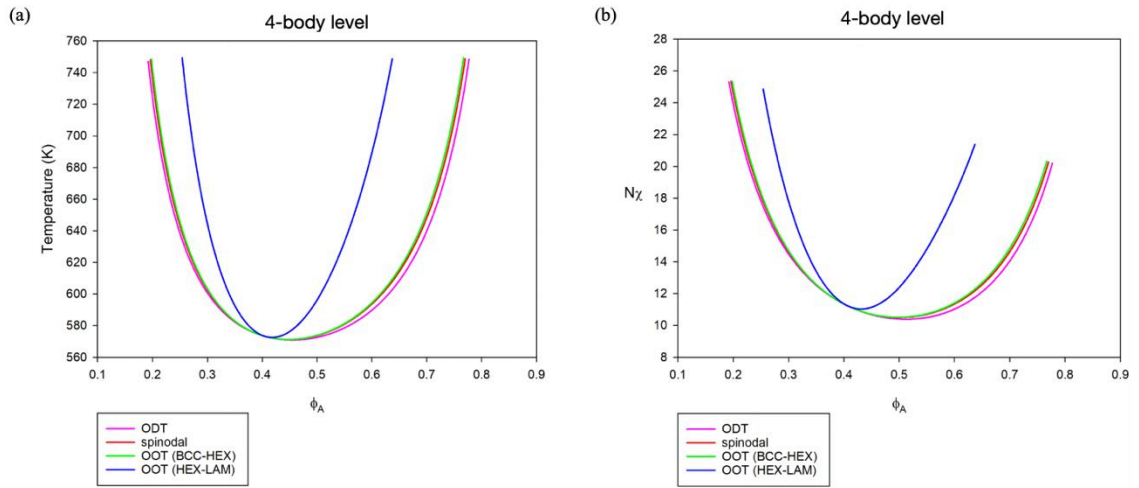


Figure 4-13. Phase diagram of PS-b-PnPrMA with $N_C = 18000$ at the four-body level, plotted as a function of ϕ_A : (a) absolute temperature and (b) the relevant parameter $N_C \chi$.

4.4 Summary

This study extends the Landau approach to associating AB diblock copolymers, based on RPA theory and a compressible Cho–Sanchez EOS. Temperature-responsive behaviors were examined using PS-*b*-PBD and PS-*b*-PVME as model systems. PS-*b*-PBD exhibits a typical UODT behavior, with nearly symmetric phase behavior due to the closely matched self-dispersion interactions of its two blocks. In contrast, PS-*b*-PVME exhibits LDOT behavior, and the large disparity in block self-dispersion interactions leads to skewed phase boundaries favoring the more compressible PVME-rich composition. Pressure-responsive behaviors were examined using PS-*b*-PBD and PS-*b*-PEHA. Upon pressurization, PS-*b*-PBD displays an increasing trend characteristic of barotropicity, whereas PS-*b*-PEHA exhibits the opposite trend of baroplasticity. These behaviors reflect the shift in dominance between enthalpic and entropic contributions under compression. Furthermore, this study further investigated the influence of multibody interactions on phase boundaries, revealing negligible effects on UODT-type diblock copolymers but significant effects on LDOT-type systems.

V. Conclusion

To gain a deeper understanding and predictive capability of the complex phase behaviors in interacting polymeric mixtures, this study systematically investigated three representative systems—polymer/ILs mixtures, polyelectrolyte solutions, and associating AB diblock copolymers—focusing on their temperature- and pressure-responsive phase transitions.

In the polymer/ILs mixtures, combined experimental characterization and molecular dynamics (MD) simulations revealed a loop-type phase behavior in the PVP/BMIMBF₄ mixture. Thermally induced conformational transitions of PVP chains, in conjunction with changes in free volume, account for the mechanism underlying this dual thermal responsiveness. Furthermore, variation of the alkyl chain length in imidazolium-based ionic liquids revealed that longer alkyl chains induce phase separation at lower temperatures and narrow ordered phase domains, demonstrating that tuning intermolecular interactions provides an effective means to regulate macroscopic phase behavior. In the polyelectrolyte solutions, a theoretical framework was developed by integrating the RPA with the compressible Cho–Sanchez EOS to construct a Landau approach for them. This approach incorporates excluded volume effects, chain connectivity, dispersion interactions, and Coulomb interactions within a unified Landau free energy form. The resulting analysis captured both ODT and OOT, revealing that chain-size asymmetry and disparities in self-dispersion interactions give rise to enriched phase diagrams, including the emergence of double CPs. The skewness of the phase diagram arises from differences in charge distribution. Barotropic and baroplastic pressure responses were also identified, revealing the enthalpy–entropy competition underlying these complex thermodynamic behaviors. The Landau theoretical framework was further extended to associating AB diblock copolymers by considering excluded volume, dispersion interactions, and the effective Flory interaction parameter, enabling investigation of the temperature- and pressure-dependent phase behaviors of various diblock copolymers. The results demonstrated that dispersion interaction asymmetry governs the skewing of phase boundaries and identified characteristic barotropic and baroplastic pressure responses. Incorporation of multibody

interactions markedly alters the LDOT phase boundaries, demonstrating that pressure-dependent compressibility amplifies density-dependent interactions, thereby coupling microscopic density fluctuations with macroscopic phase behavior in associating AB diblock copolymers.

In conclusion, this work provides a theoretical foundation for predicting the complex phase behaviors of interacting polymeric systems and elucidates the underlying molecular mechanisms, offering new insights for the rational design of smart materials.

Appendix. SCFT-Based Visualization of Block Copolymer Nanostructures Using 3D Printing

Block copolymers self-assemble into ordered nanoscale morphologies that can serve as templates for functional materials, yet direct visualization of these structures is often challenging. The emergence of 3D printing has provided a means to visualize and demonstrate these complex morphologies. This appendix describes the methodology for visualizing block copolymer nanostructures by integrating SCFT with 3D printing.¹⁴¹ In this study, the method of designing a model directly by 3d design software is used to get a simple ordered structure model. For complex morphologies, quantitative morphology data obtained from SCFT, including nanostructure viscosity information, were further converted into G-code (XYZ format) to define the triangular surface mesh. The resulting mesh was imported into 3D design software to generate STL files, which were subsequently used for 3D printing, producing tangible physical models of the predicted nanostructures.

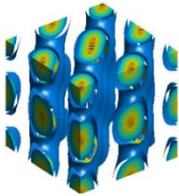
Figure A1–Figure A5 illustrate the visualization results for representative block copolymer nanostructures, including Im3, double gyroid (Ia $\bar{3}$ d), Metatron cube, hexagonal (P6mm), and C2mm morphologies.

For each structure:

- (a) shows the SCFT-predicted morphologies obtained from numerical simulations performed with MATLAB or Materials Studio;
- (b) presents the 3D models constructed using 3D design software;
- (c) displays the 3D-printed nanostructures fabricated based on the generated STL files;
- (d) depicts the corresponding scattering peak analyses.

Im3

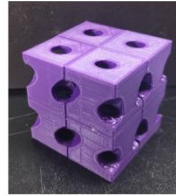
110 N_A
 55 N_B
 55 N_C
 Box size=5.007*5.007*5.007



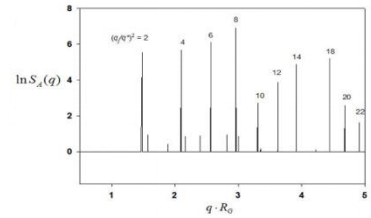
(a)



(b)



(c)

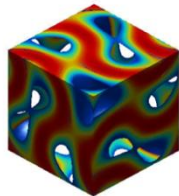


(d)

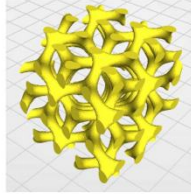
Figure A1. Visualization of the Im3 morphology: (a) SCFT-predicted morphology obtained from numerical simulations using MATLAB; (b) 3D model generated by 3D design software; (c) 3D-printed nanostructure; (d) corresponding scattering peak analysis.

Double gyroids

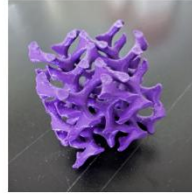
110 N_A
 55 N_B
 55 N_C
 Box size=8.727*8.747*8.727



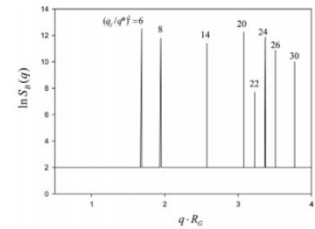
(a)



(b)



(c)

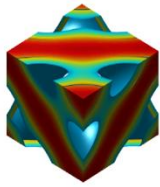


(d)

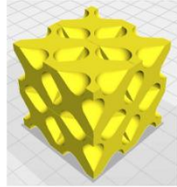
Figure A2. Visualization of the double gyroid morphology: (a) SCFT-predicted morphology obtained from numerical simulations using MATLAB; (b) 3D model generated by 3D design software; (c) 3D-printed nanostructure; (d) corresponding scattering peak analysis.

Metatron cube

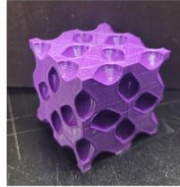
110 N_A
 55 N_B
 55 N_C
 Box size=5.304*5.304*5.304



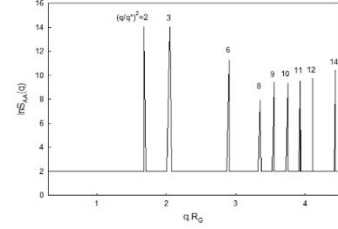
(a)



(b)



(c)

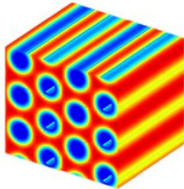


(d)

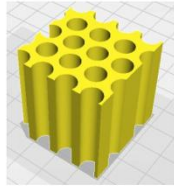
Figure A3. Visualization of the Metatron cube morphology: (a) SCFT-predicted morphology obtained from numerical simulations using MATLAB; (b) 3D model generated by 3D design software; (c) 3D-printed nanostructure; (d) corresponding scattering peak analysis.

Hexagonal P6mm

110 N_A
 55 N_B
 55 N_C
 Box size=4.064369*7.039694*10



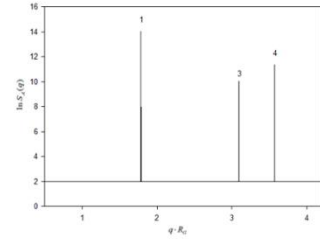
(a)



(b)



(c)

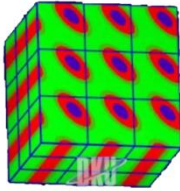


(d)

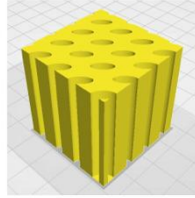
Figure A4. Visualization of the hexagonal (P6mm) morphology: (a) SCFT-predicted morphology obtained from numerical simulations using MATLAB; (b) 3D model generated by 3D design software; (c) 3D-printed nanostructure; (d) corresponding scattering peak analysis.

C2mm

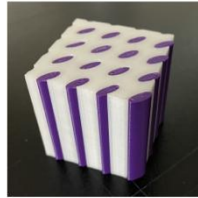
110 N_A
 55 N_B
 55 N_C
 Box size=4.717*4.717*4.717



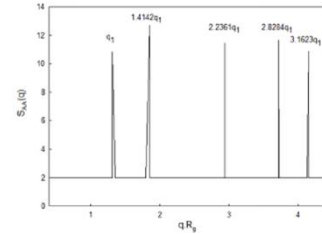
(a)



(b)



(c)



(d)

Figure A5. Visualization of the C2mm morphology: (a) SCFT-predicted morphology obtained from numerical simulations using Materials Studio; (b) 3D model generated by 3D design software; (c) 3D-printed nanostructure; (d) corresponding scattering peak analysis.

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(국문초록)

인력 작용 하 고분자 혼합물의 복잡 상거동:

열역학적 해석 및 분자동역학 시뮬레이션

단국대학교 대학원 고분자공학과

고분자물리전공

자오 밉 거

지도교수 : 조 준 한

인력 작용 하의 고분자 혼합계는 분자 간 인력의 다양성과 조절 가능성으로 인해 자극 반응형 소재를 설계하기 위한 핵심 연구 대상으로 주목받고 있다. 그러나 이러한 시스템의 상거동은 다중 스케일 인력 작용의 경쟁과 결합에 의해 지배되며, 그 복잡성이 높아 이론적 해석과 실험적 예측에 상당한 어려움을 초래한다. 본 연구에서는 이러한 문제를 해결하기 위하여 고분자/이온성 액체 혼합계, 고분자전해질 블렌드, 회합성 블록 공중합체의 세 가지 대표 시스템을 대상으로 다중 스케일 인

력 작용 하에서의 상거동 조절과 미세구조의 열역학적 진화 메커니즘을 규명하였다. 이를 통해 스마트 소재 설계를 위한 체계적이고 일반화된 이론적 틀을 제시하고자 하였다.

첫째, 고분자/이온성 액체 혼합계의 우수한 열반응성에 주목하여, polyvinylpyrrolidone (PVP)/1-butyl-3-methylimidazolium tetrafluoroborate (BMIMBF₄) 시스템의 상거동을 실험적 분석과 분자동역학 (MD) 시뮬레이션을 통합적으로 연구하였다. 가열 과정에서 시료는 투명-불투명-투명의 전이 현상을 나타냈으며, 이는 DSC와 광산란 분석을 통해 온도 및 고분자 농도 의존성을 정량적으로 분석하였다. 본 시스템은 LCST (lower critical solution temperature)와 UCST (upper critical solution temperature)가 동시에 존재하는 한 독특한 루프형 (loop-type) 상전이 거동을 나타냈으며, 이는 기존의 단일 열반응성 시스템보다 상전이 조절의 유연성과 적응성이 뛰어남을 보여주었다. 분자 수준의 메커니즘을 밝히기 위해 OPLS5 역장 (force field)를 적용한 MD 시뮬레이션을 수행하였다. 구조 및 군집 (cluster) 분석 결과, 가열 과정에서 PVP 사슬이 점진적 붕괴에서 재신장으로 전이되는 열유도 구조 변화를 나타냈으며, 자유 부피 분석을 통해 이러한 루프형 상거동이 더욱 검증되었다. 또한, 이미다졸리움 (imidazolium) 기반 이온성 액체의 알킬 사슬 길이를 변화시킨 시뮬레이션을 통해, PVP 사슬은 온도에 따른 유사한 열유도 구조 변화를 나타내었으며, 이는 이러한 분자적 특성이 PVP/이온성 액체 혼합계의 루프형 상거동을 효과적으로 제어함을 보여주었다. 이러한 발견은 열반응성 스마트 소재 설계를 위한 분자 수준의 새로운 통찰을 제공한다.

둘째, 고분자전해질 블렌드에서는 정전기적 상호작용과 사슬 구조 간의 상호작용이 복잡하고 다양한 상거동을 유발한다. 상반된 전하를 가진 고분자전해질 블렌드의 상거동을 체계적으로 분석하기 위해, 하전계를 위한 상태방정식 (Cho-Sanchez E

OS for charged systems)과 결합한 Random-phase Approximation (RPA) 이론에 기반한 Landau 자유 에너지를 구축하였다. 이 이론적 틀은 배제체적 효과, 사슬 연결성, 분산 상호작용, 장거리 쿨롱 인력을 모두 포함한다. 본 모델을 활용하여, 기존의 질서-무질서 전이뿐만 아니라, 완전 충전부터 약하게 충전된 시스템에 이르는 보다 복잡한 질서 전이도 함께 분석하였다. 사슬 길이 비대칭, 전하 효과, 분산 상호작용 격차 및 압력 효과가 상거동 형성에 중요한 역할을 함을 확인하였다. 또한, 질서 전이에서 나타나는 압방성 (barotropicity) 및 압가소성 (baroplasticity)을 규명하여, 압력에 응답하는 고분자전해질 소재 설계 가능성을 강조하였다. 이러한 연구 결과는 고분자전해질 시스템의 이론적 예측을 위한 메커니즘적 이해를 제공하고, 시스템의 합리적 설계를 위한 일반 원리를 제시한다.

셋째, 본 연구의 이론적 틀은 회합성 AB 블록 공중합체로 확장되었다. 회합성계를 위한 상태방정식 (Cho-Sanchez EOS for associating systems)과 결합한 RPA 이론에 기반하여, 배제체적 효과, 분산 상호작용, 유효 Flory 상호작용 파라미터 (χ ; 엔탈피 기여 χ_H 와 엔트로피 기여 χ_S 의 합으로 정의)를 고려한 Landau 자유 에너지 함수를 구성하였다. RPA 분석을 통해 임계 조건을 규명하고 유효 상호작용 파라미터를 정량화하였다. 자기충족적 장 이론(SCFT)을 통해 평형형 및 구조 진화를 기술하고, 다양한 블록 공중합체에서 나타나는 온도 및 압력 의존적 질서 형성을 규명하였다. 특히, 엔탈피와 엔트로피 기여 간의 경쟁이 냉각 유도 질서뿐만 아니라, 상대적으로 드문 가열 유도 질서를 유발함을 확인하였다. 편향된 상경계 (skewed phase boundary) 는 분산 상호작용 비대칭성이 질서형 구조 형성에 중요한 역할을 함을 강조한다. 더 나아가, 압방성 및 압가소성 응답을 규명하고, 다체 상호작용 (multibody interaction)이 상경계에 미치는 영향을 추가적으로 탐구하여, 회합성 블록 공중합체의 자기조립 메커니즘에 대한 이해를 확장하였다. 이러한 연구 결과는 열 감응성과 압력

응답성을 갖춘 나노구조화 소재의 합리적 설계 및 개발을 위한 이론적 지침을 제공한다.

감사의 글

이 감사의 글을 쓰는 지금, 저의 대학원 생활이 마무리되어 가고 있음을 실감합니다. 만남과 이별이 교차하는 시간 속에서 지난날을 돌아보면, 제 생각과 학문 모두 한층 성숙해졌음을 느낍니다. 모든 경험은 제게 소중한 선물이었고, 모든 인연은 값진 보물이었습니다. 처음 설렘을 안고 한국에 발을 디딘 순간부터, 박사 논문을 완성하기까지의 여정은 한 편의 연구 여정이자 자기 성찰의 시간들이었습니다. 이 긴 여정 속에서 제가 성장할 수 있었던 것은 수많은 분들의 도움과 따뜻한 동행 덕분이었습니다.

먼저, 진심으로 지도교수님 조준한 교수님께 깊은 감사의 말씀을 드립니다. 교수님께서는 언제나 학문에 대한 열정과 엄격한 연구 태도로 저를 이끌어 주셨습니다. 세심한 지도와 따뜻한 격려 덕분에 연구의 방향을 잡고 학문적으로 한 단계 성장할 수 있었습니다. 교수님과 함께한 시간은 제 연구 인생에서 가장 값지고 뜻 깊은 배움의 여정이었습니다.

또한, 논문 심사와 연구 과정에서 귀중한 조언을 아끼지 않으신 여러 교수님들께도 깊이 감사드립니다. 교수님들께서 주신 소중한 말씀은 연구 주제를 더욱 깊이 이해하고, 과학적 탐구가 지녀야 할 넓이와 깊이를 배우는 계기가 되었습니다. 아울러, 강의와 연구를 통해 저에게 큰 영감과 배움을 주신 이동현 교수님, 김세영 교수님, 오준균 교수님, 김응건 교수님, 강호종 교수님, 황석호 교수님, 진병두 교수님, 구병진 교수님께 진심으로 감사의 말씀을 드립니다. 풍부한 지식과 독창적인 통찰, 그리고 열린 사고를 바탕으로 한 가르침은 제게 학문적 열정과 탐구의 방향을 제시해 주셨습니다.

실험실에서 함께 연구하고 생활하며 서로에게 큰 힘이 되어 준 친구들과 동료들에게 깊은 감사의 마음을 전합니다. 연구와 일상 속에서 늘 따뜻한 격려와 도움을 아끼지 않은 장신위에, 김신철 후배에게 진심으로 감사드립니다. 특히, 한국에서의 생활과 성장 과정에서 많은 조언과 정성 어린 도움을 베풀어 준 정수욱에게도 깊이 감사드립니다. 힘든 순간마다 함께 고민하고 웃으며 보낸 시간들은 제게 큰 위로와 용기가 되었고, 여러분 덕분에 낯선 외국 생활이 외롭지 않았으며 매일의 연구가 더욱 의미 있게 느껴졌습니다.

멀리서 늘 믿어 주시고 응원해 주신 부모님께 가장 깊은 감사를 드립니다. 부모님의 사랑이 있었기에 저는 이 길을 끝까지 걸어올 수 있었습니다.

마지막으로, 지금까지 포기하지 않고 묵묵히 걸어온 자신에게도 감사의 마음을 전하고 싶습니다. 수많은 밤과 낮 속에서 흔들리며 끝내 버텨 낸 시간들이 저를 성장시켰습니다. 그 과정 속에서 불확실함을 마주하는 법을 배우고, 시간과 노력의 힘을 믿게 되었습니다. 박사 과정의 여정은 끝이 아닌 새로운 시작이라 생각합니다. 앞으로도 초심을 잃지 않고 더 넓은 세상 속에서 꾸준히 탐구하며 나아가길 바랍니다.

2025 년 12 월
자오밍거